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Term 7- Sept 2025

Nanoelectronics and Technology (01.119/99.503)-Week 13 Class 1

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09-Dec- 2025

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Outline

- Introduction to reliability and Yield of electronics devices
- Importance of reliability of MOSFET devices with CMOS scaling
- Introduction to MOSFET reliability
 - Overview of gate dielectric scaling
 - Generation of defects
 - Stages of gate dielectrics degradation
 - Dielectric Breakdown mechanisms and models

Reliability of Electronic Devices

- Reliability is a measure of the frequency of equipment failures as a function of time. Reliability has a major impact on maintenance and repair costs and on the continuity of service.
- A practical definition of reliability is “the probability that a piece of equipment operating under specified conditions shall perform satisfactorily for a given period of time”.
- MTTF (mean time to failure) applies to parts that will be thrown away on failing.

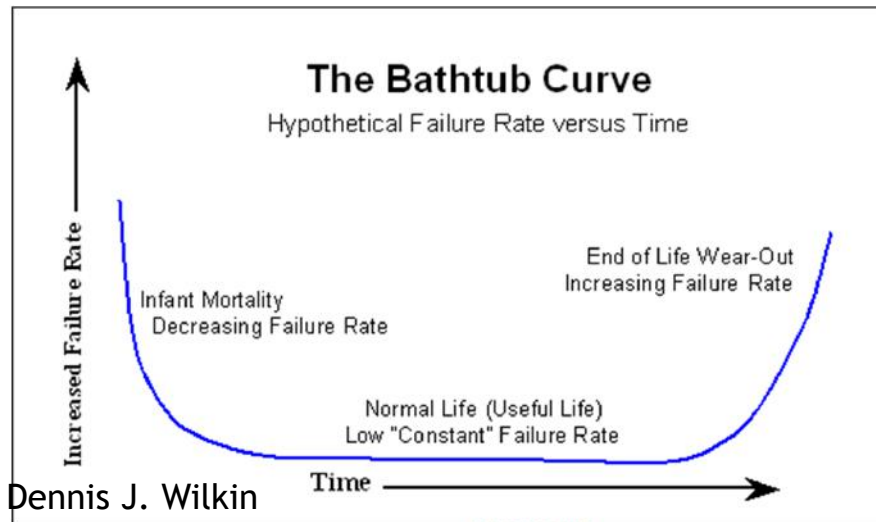
Reliability of Electronic Devices

Factors affecting Reliability

- Design factors
- Complexity
- Stress
- Inherent reliability of materials

Reliability and its impact on failure

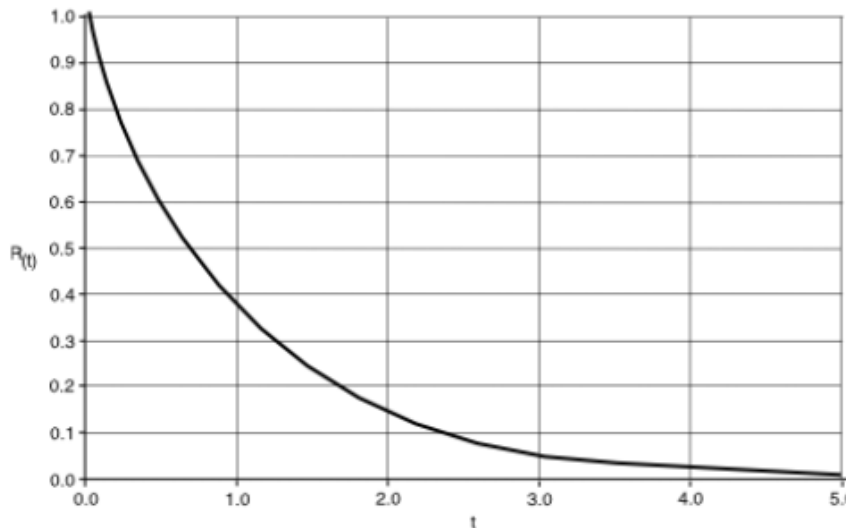
- Failure rate is defined as the percentage of units failing per unit time. This varies throughout the life of the equipment and plotted against time
- The characteristic “bathtub” curve is obtained for most electronic equipment



<https://www.weibull.com/hotwire/issue21/hottopics21.htm>

Reliability and its impact on failure

- Wear out of electronic devices different from mechanical devices.
- There are debatably some wear out mechanisms such as component parameter drift and electromigration. Electronic systems behave fundamentally different than mechanical ones.



Reliability $R(t)$ Plotted Against (λt) for a unit with a constant failure rate λ

Reliability and its impact on failure

Estimating the failure rate: The failure rate should be estimated and measured throughout the life of the equipment

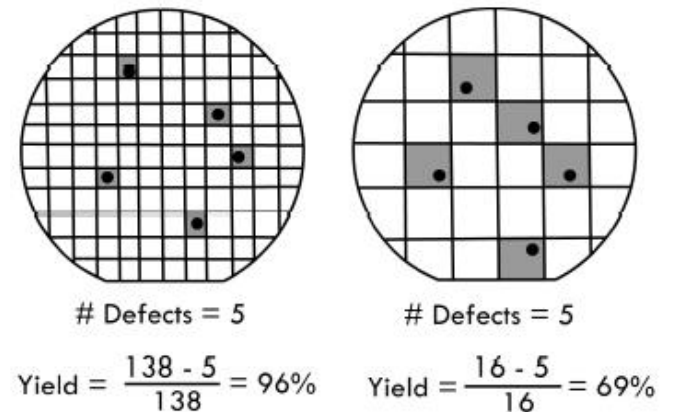
- During Design, it is predicted
- During manufacturing, it is assessed
- During the service, it is observed

Yield

Device yield or die yield is **the number of working chips or dies on a wafer**, given in percentage since the number of chips on a wafer (Die per wafer) can vary depending on the chips' size and the wafer's diameter.

$$\text{Yield} = \frac{\text{Number of good (working) chips on wafer}}{\text{Total number of chips}}$$

- Yield is influenced by
 - The technology, chip area, layout

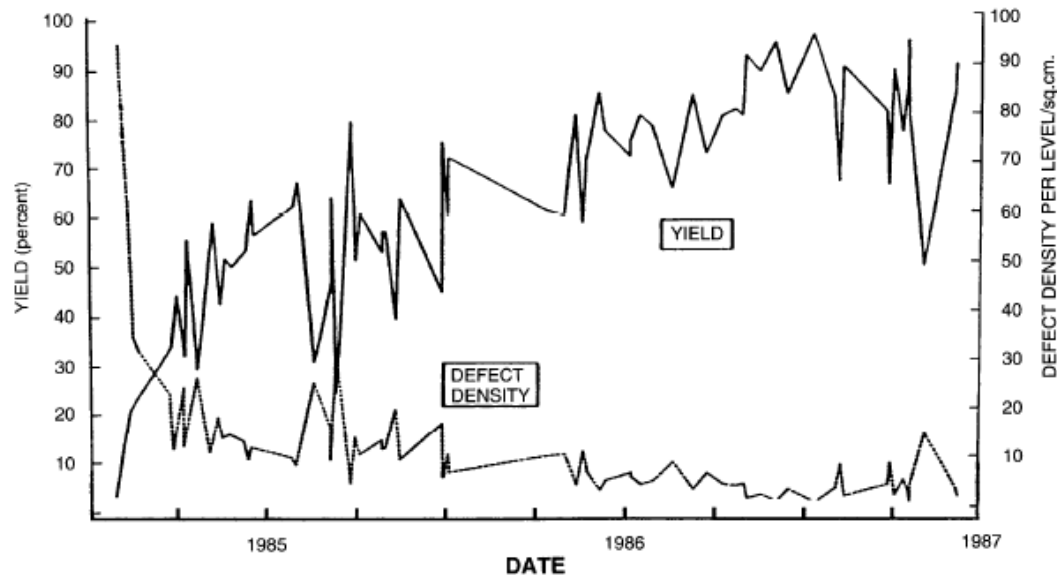


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Defects/Failures and its impact on Yield

- DEFECTS cause FAILURES.
- Failures caught during testing or screening lead to **YIELD loss**.
- Latent defects which escape product testing could lead to FAILURES during service –RELIABILITY loss.

YIELD vs RELIABILITY – defect is the common link



Defect density and yield variations produced when monitoring the impact of yield enhancement actions in a 1.25 μm process.
From J. M. Pimbley, M. Ghezzi, H. G. Parks, and D. M. Brown, in *VLSI Electronics Microstructure Science*, Vol 19, Academic Press (1989)

Yield is Measured.

Reliability is Predicted.

Yield is mainly a concern of the manufacturer

Reliability is mainly a concern of the user.

Reliability and its impact on failure

- **Wafer Size:** Larger wafer size certainly provides more die per unit area. The extra area to fabricate more die eventually allows FABs to fabricate and test/assemble more dies in a given time. This pushes the rate at which new products can be fabricated/assembled and to some extent increasing wafer size can also have a positive impact on the supply chain.
- **Die Per Wafer:** Wafer size clearly defines how many die per wafer there will be. This allows the semiconductor design houses to gauge how much cost savings will be there.

Reliability and its impact on failure

- **Wafer Size:** Larger wafer size certainly provides more die per unit area. The extra area to fabricate more die eventually allows FABs to fabricate and test/assemble more dies in a given time. This pushes the rate at which new products can be fabricated/assembled and to some extent increasing wafer size can also have a positive impact on the supply chain.
- **Die Per Wafer:** Wafer size clearly defines how many die per wafer there will be. This allows the semiconductor design houses to gauge how much cost savings will be there.
- **Cost:** Wafer size certainly dominates the cost of developing a semiconductor product. Apart from the cost of the wafer itself, there are FAB costs that also need to be considered. Using a 200 mm (7.9/8 inch) wafer will certainly have a lower cost of fabricating and assembling semiconductor chips compared to a 300 mm (11.8/12 inch) wafer. In the end, it is all about creating the margin by selecting the right wafer size.

Yield

In general:

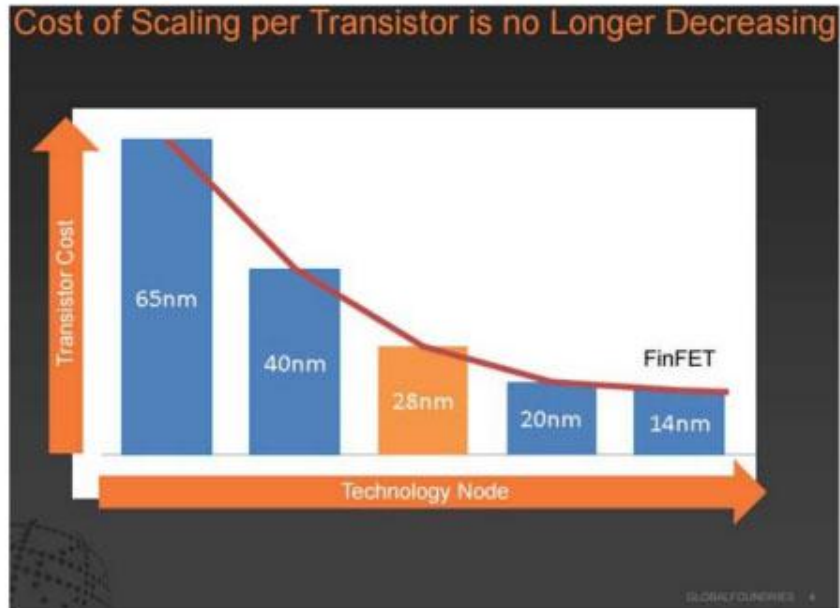
- Yield drops as the chip area increases
- For a given chip area, higher yields for lower defect density
- For larger chips, lower defect density is necessary to maintain the same yield

Yield

Unfortunately, larger chips and reduced dimensions imply the following:

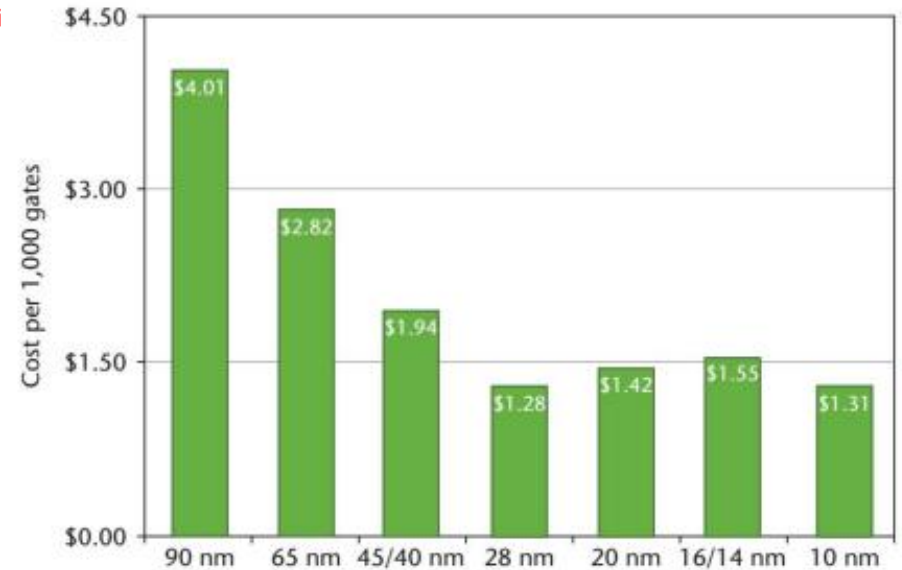
- Critical defect size is smaller –(narrower lines can be destroyed by a smaller particle) –a higher defect *sensitivity*
- Larger chips have larger exposure to process defects
- In addition, larger chips are usually accompanied by higher process complexity (more steps), thus increasing the probability of incorporating defects.

Cost factor



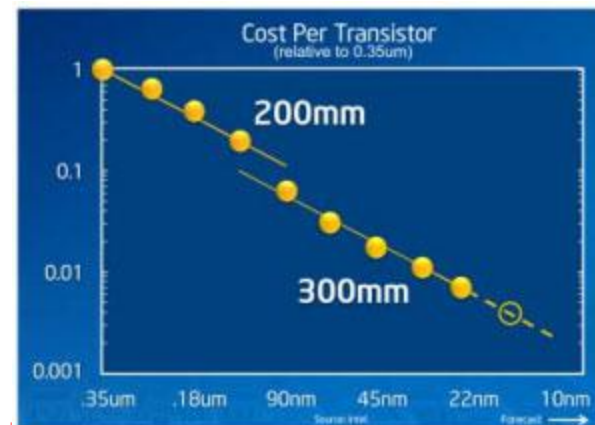
Global Foundries' Transistor Manufacturing Cost at Recent Technology Nodes
Source: McCann (2015).

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Cost per logic gate, with projection for 10nm technology node

Source: Jones (2015)



Source: Jones

<https://www.nber.org/papers/w24553>

Why Reliability study and Failure analysis required?

- As device dimensions shrink into the nanometer regime, reliability and failure analysis become mandatory, not optional.
- Modern transistors (FinFETs, GAA nanosheets, 3D NAND, RRAM, BEOL low-k) operate at extreme electrical, thermal, and material limits. Without reliability engineering, chips would fail within days or even hours.
- At nanoscale, even minor changes in material properties and operating conditions can shift V_{th} .
- Reliability is linked to yield.
- Variability leads to parametric failures.
- early failures reduce production efficiency.
- Foundries rely on reliability analysis to control variation.

Why Reliability study and Failure analysis required?

Reliability and failure analysis are crucial in nanoelectronics because:

- Devices are extremely small → very sensitive to defects.
- New failure mechanisms appear at nanoscale.
- Electric fields and current densities are extremely high.
- Variability makes device behavior unpredictable.
- Billions of devices on one chip → one failure causes system failure.
- Applications require long life and high safety.
- Only failure analysis reveals what actually kills devices.

Why Reliability study and Failure analysis required?

As devices shrink:

- current density $\uparrow$
- power density $\uparrow$
- heat dissipation $\downarrow$

Self-heating leads to:

- faster BTI
- local thermal runaway
- dielectric breakdown hotspots
- mobility degradation

Failure characteristics

Stresses

Operational variables that are related to known failure mechanisms.

Using them above specified usage conditions constitutes **Acceleration**.

- **Voltage (Current) stress**

Dielectric breakdown, electromigration, hot-carrier damage

Time to failure for oxide breakdown and hot-carrier damage decreases exponentially as voltage stress is increased.

Time to failure for electromigration is inversely proportional to $\sim$ square of the current density

- **Temperature Stress**

Accelerate thermally-activated processes.

Relevant for oxide breakdown, electromigration and package-related failures.

Failure characteristics

ACCELERATION (relate rate to stress)

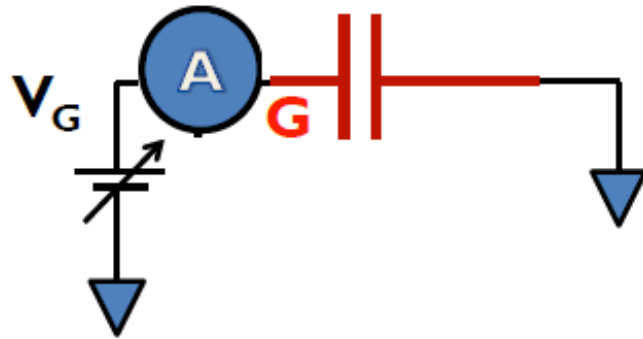
speeding up failures to reasonable times using higher-than-normal stresses. Then extrapolate to failure times at normal conditions.

Thermal Acceleration

Many chemical and physical processes are thermally activated, meaning that they may be accelerated by increasing the temperature.

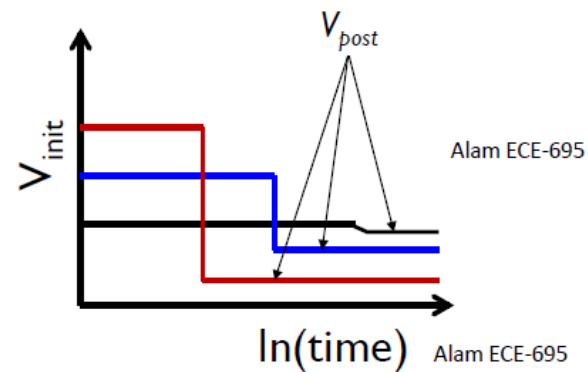
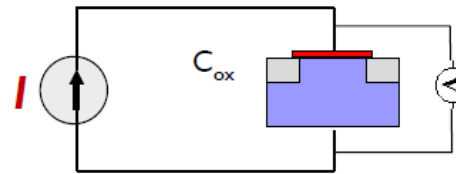
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Gate dielectric Reliability and Failure analysis



Const. Current Stress

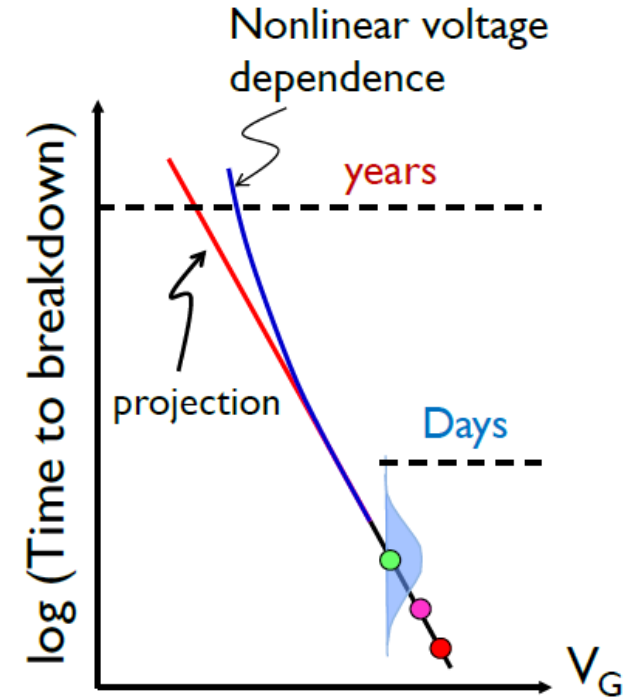
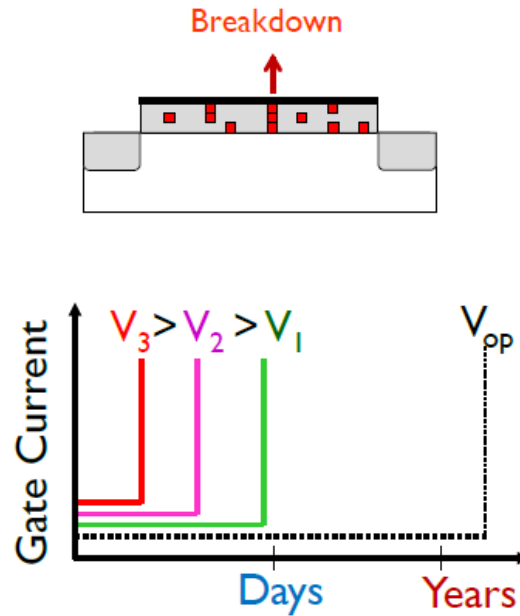
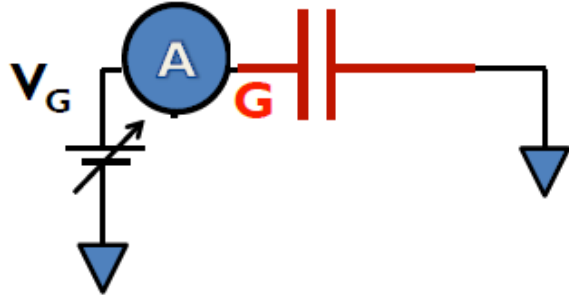
Constant Current Stress



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Gate dielectric Reliability



Highly accelerated voltage dependence

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Reliability Analysis

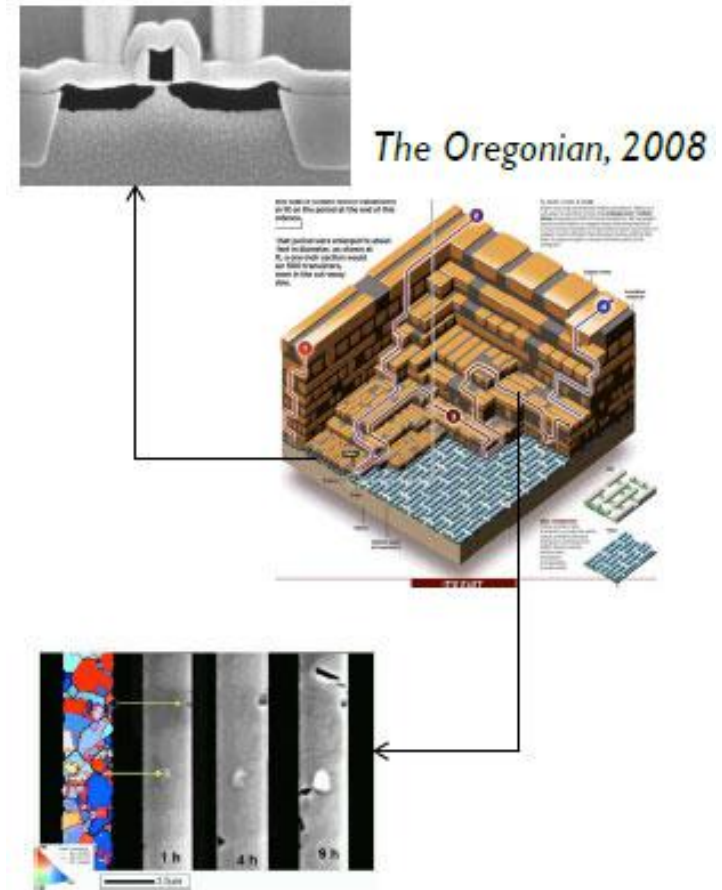
Reliability of Nanotransistors

Transistor Reliability

- Gate Dielectric Breakdown
- Hydrogen release model
- Anode hole injection model
- Hot Carrier Degradation
- Radiation Induced damage

Interconnect Reliability

- Electro-migration/Stress-migration.
- Inter-level dielectric breakdown

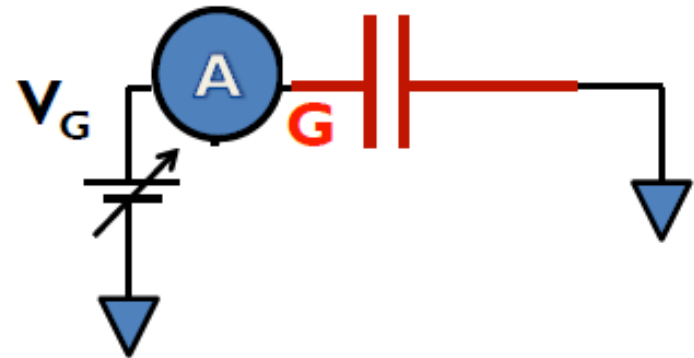
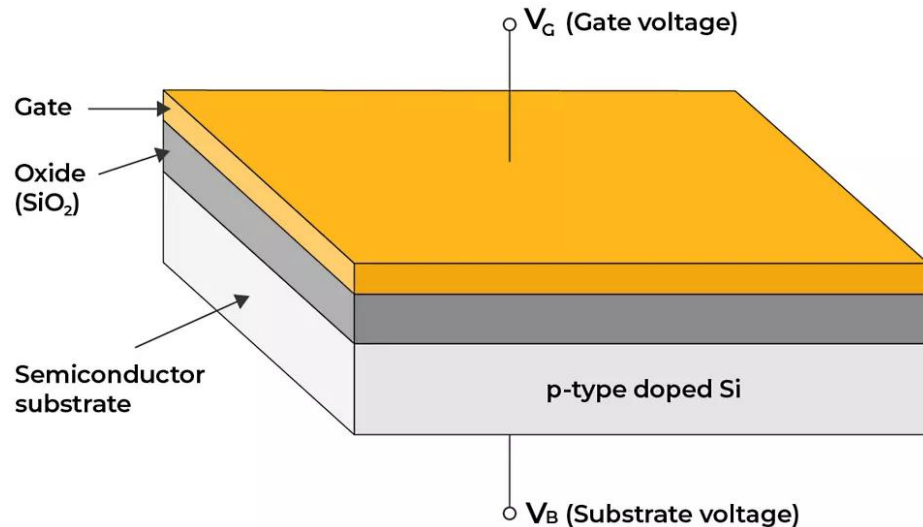


Ref: nanohub.org (M. Alam ECE 695)

Gate dielectric Reliability and Failure analysis

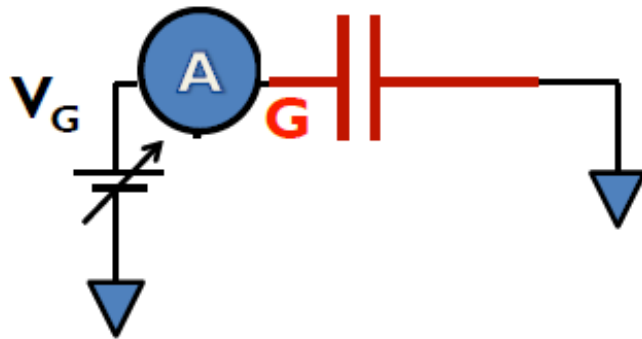
How to study reliability of the device (Dielectric)- Accelerated stressing

Accelerated Stress Testing can be simply defined as: applying high levels of stress for a short period of time to a device under test assuming it will exhibit the same failure mechanisms as it would in a longer amount of time at lower stress levels.



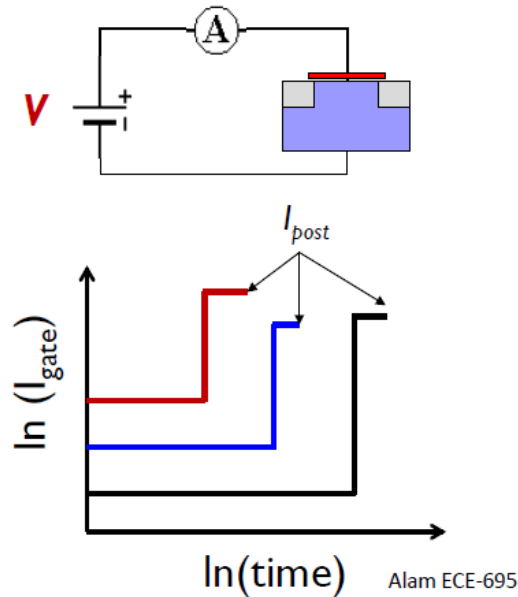
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Gate dielectric Reliability



Const. Voltage Stress

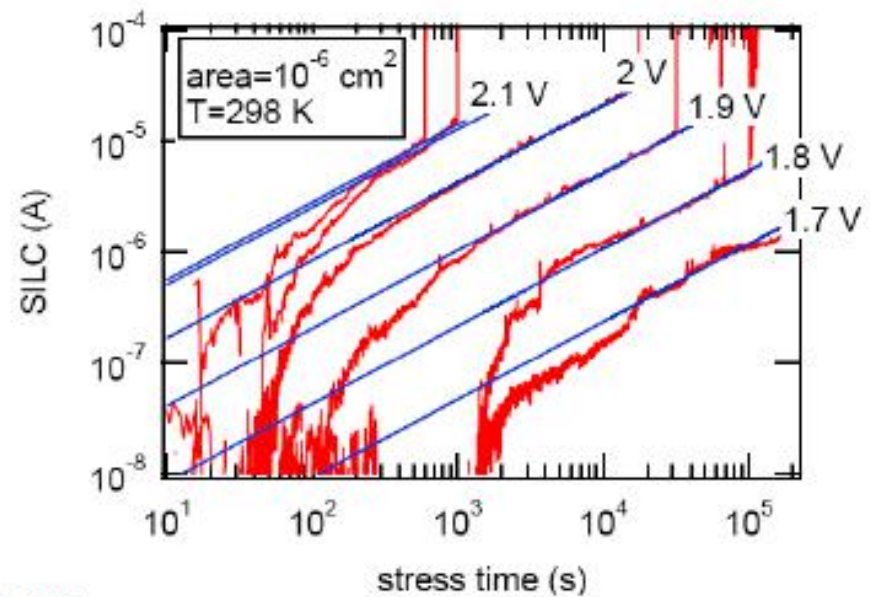
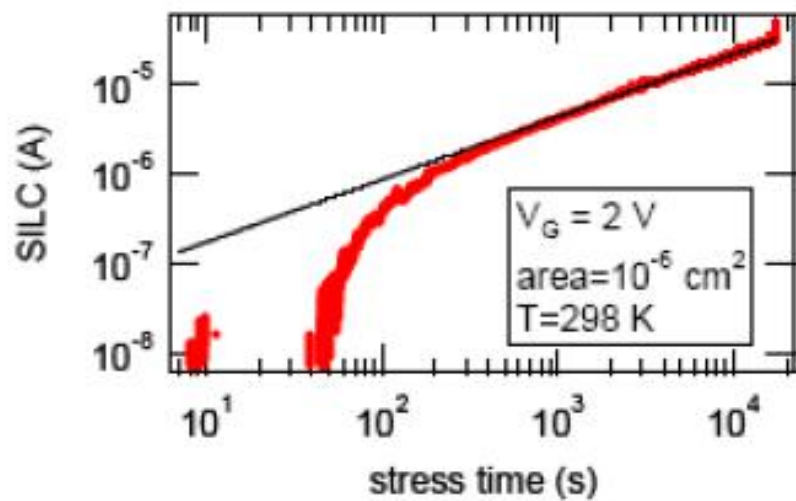
Constant Voltage Stress



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Gate dielectric Reliability and Failure analysis



Alam ECE-695

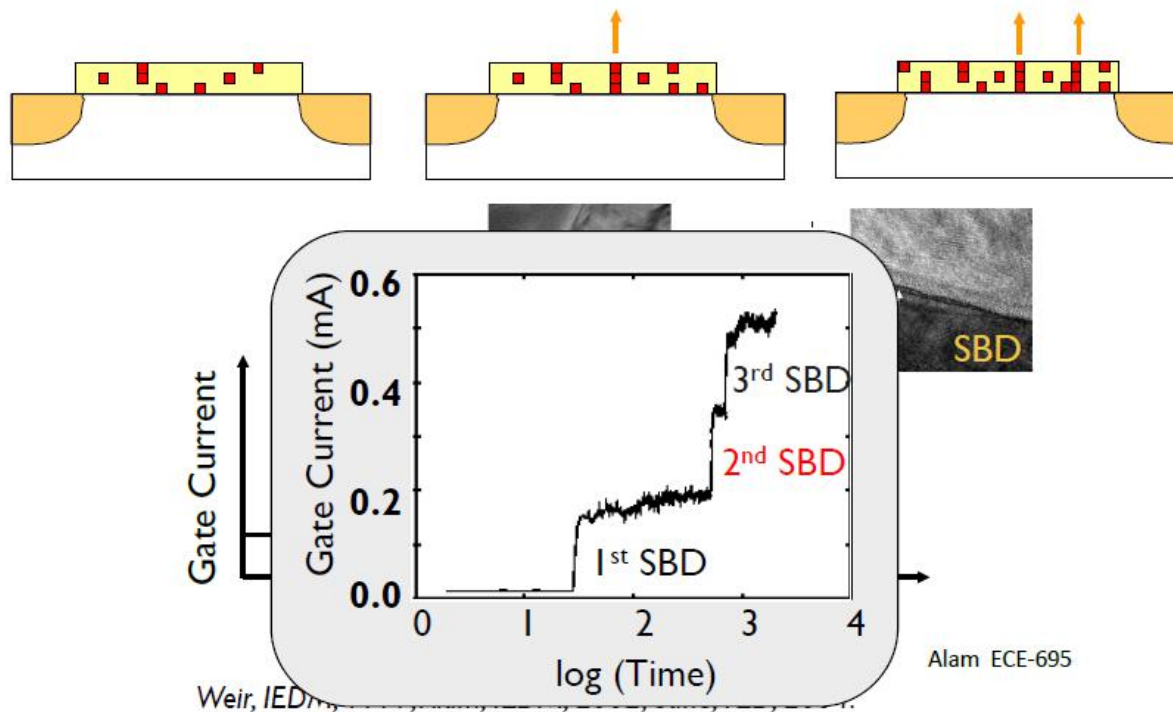
K.L. Pey, Course material - Gate oxide dielectric failure mechanisms and reliability-NTU

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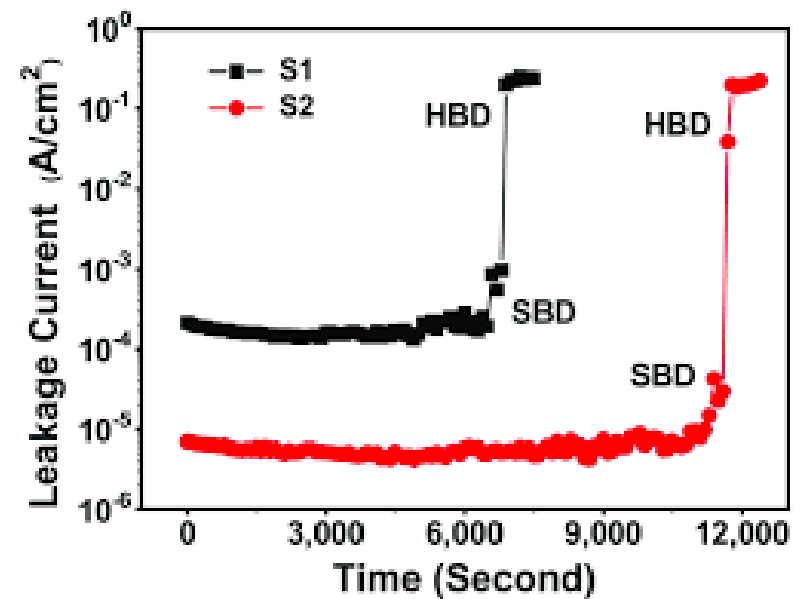
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Gate dielectric Reliability and Failure analysis

Soft breakdown



Alam ECE-695



[Materials](#) 11(9):1601, Sept 2018

ITN [10.3390/ma11091601](#)

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Gate dielectric Reliability and Failure analysis

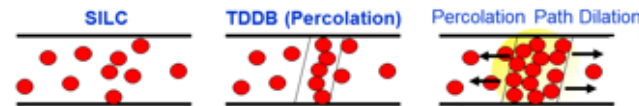
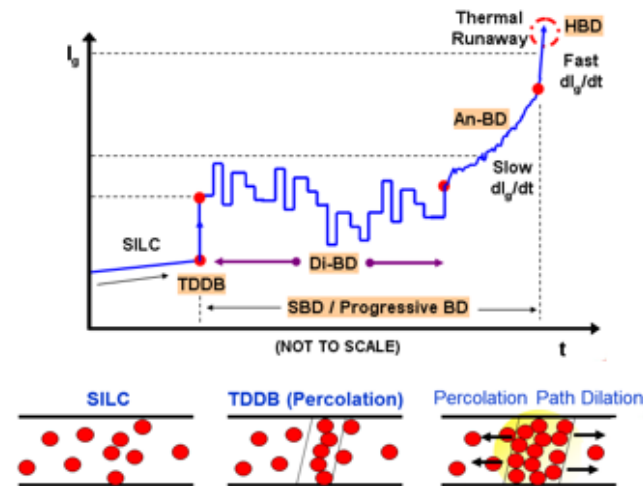
An Overall Picture of Dielectric BD Evolution

Conventional Device level BD analysis

- Stress induced leakage current (SILC)
- Time dependent dielectric breakdown (TDDB)
- Progressive breakdown and hard breakdown (HBD)

Trap generation models

- Thermo-chemical model
- Anode-hole injection model
- Hydrogen release model



N. Raghavan, PhD Thesis, NTU-2012

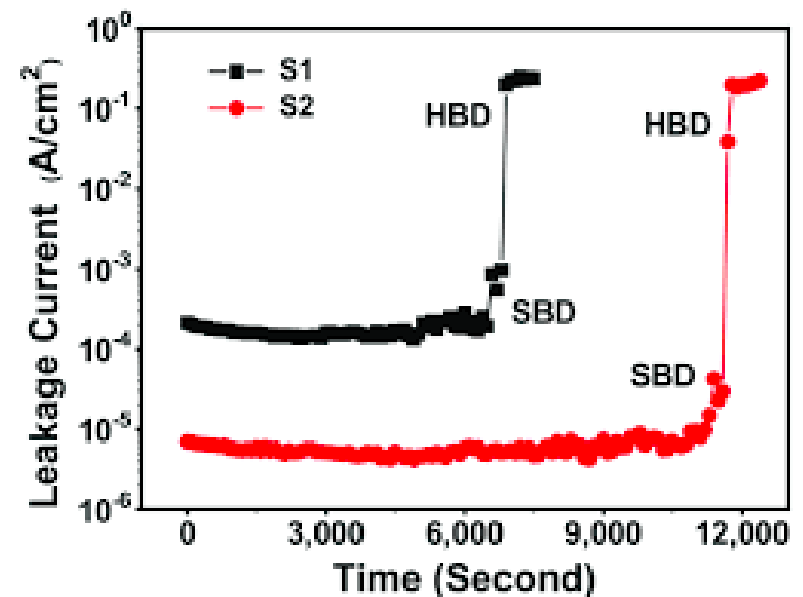
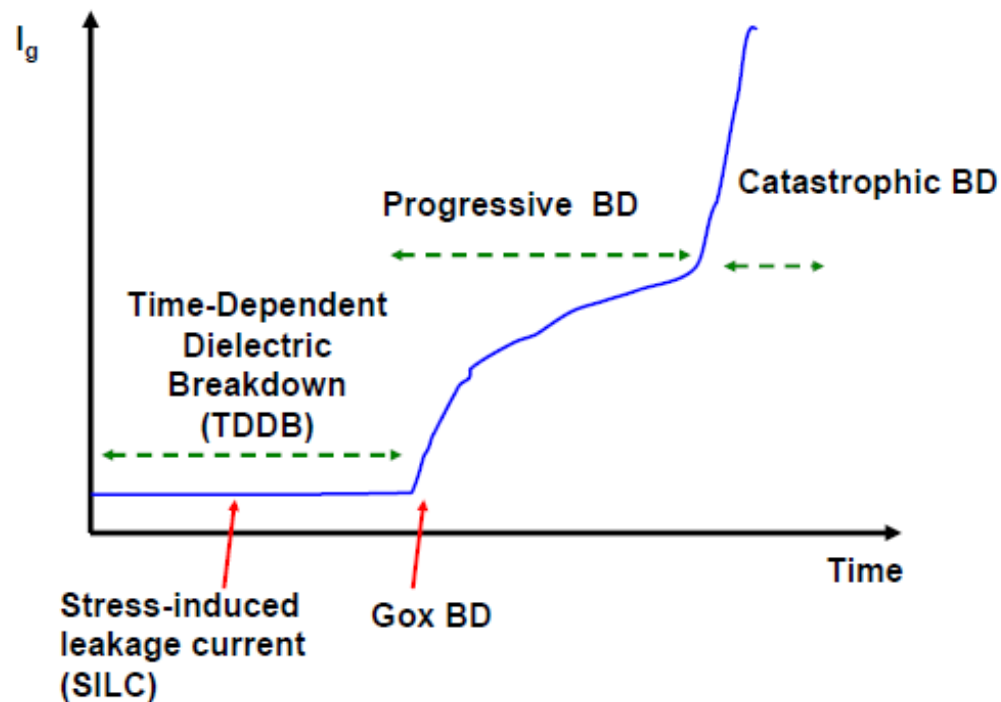
J. W. McPherson, et al., IEEE-TED, 50,1771 (2003)

M. Houssa, et al., APL, 79, 3134 (2001)

K.L. Pey, Course material - Gate oxide dielectric failure mechanisms and reliability-NTU

Gate dielectric Reliability and Failure analysis

Stages of gate dielectrics degradation wear-out & progressive breakdown

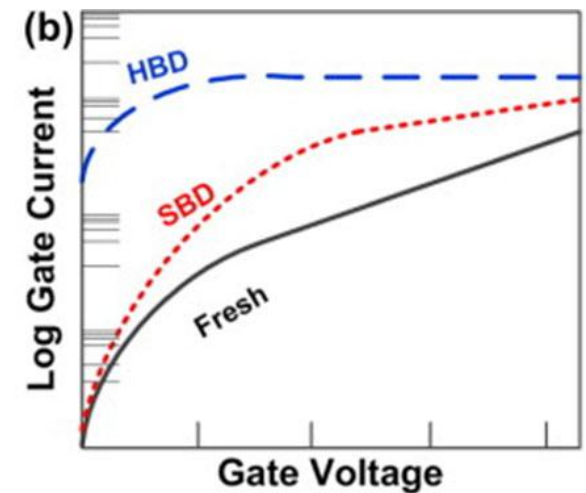
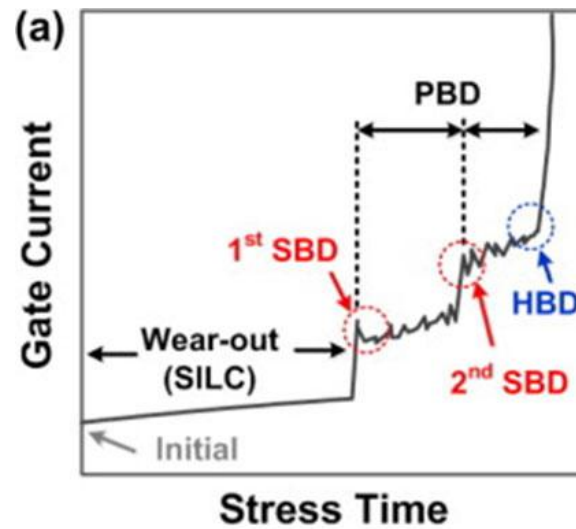


Materials 11(9):1601, Sept 2018
 ITM [10.3390/ma11091601](https://doi.org/10.3390/ma11091601)

K.L. Pey, Course material - Gate oxide dielectric failure mechanisms and reliability-NTU

Gate dielectric Reliability and Failure analysis

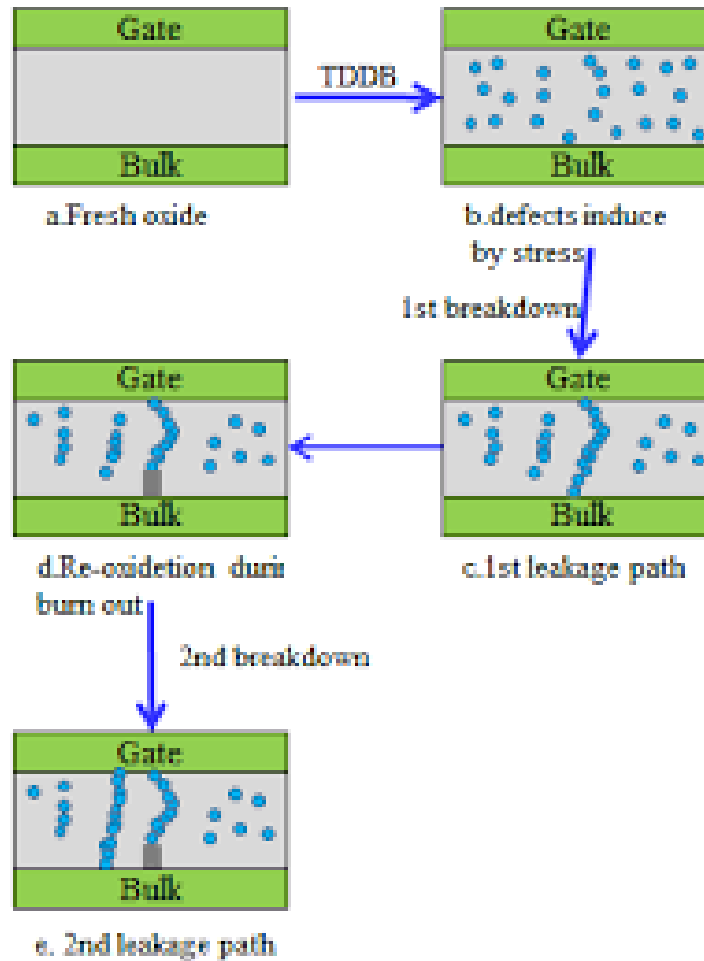
- Stress induced leakage current (SILC)
- Time dependent dielectric breakdown (TDDB)
- Progressive breakdown and hard breakdown (HBD)



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Gate dielectric Reliability and Failure analysis

- Stress induced leakage current (SILC)
- Time dependent dielectric breakdown (TDDB)
- Progressive breakdown and hard breakdown (HBD)

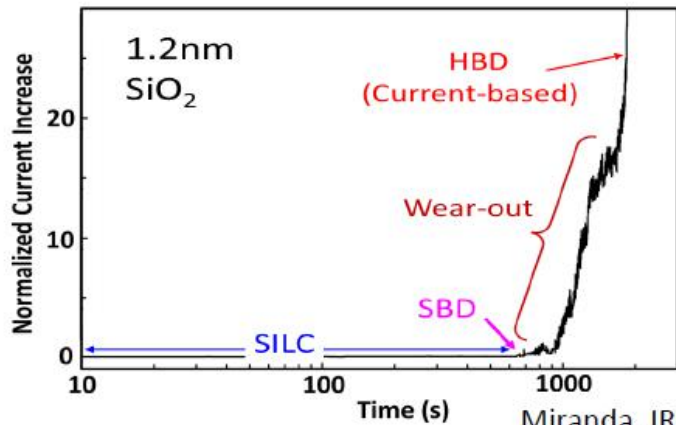


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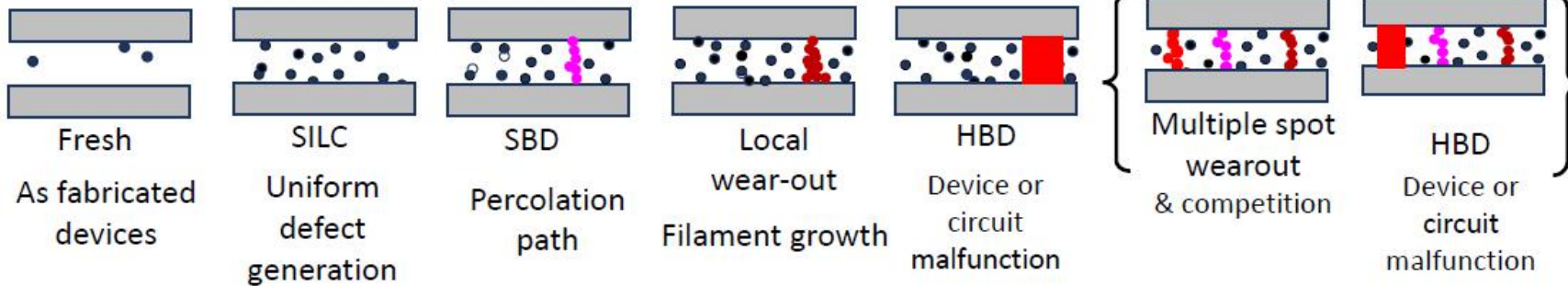
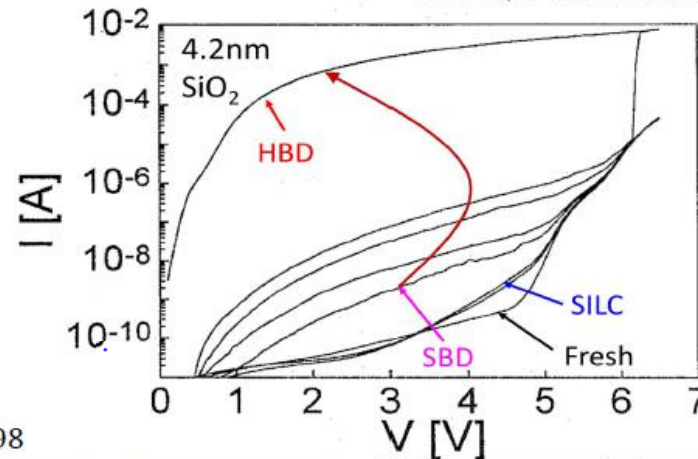
Gate dielectric Reliability and Breakdown

Various stages in defect generation and dielectric breakdown

Ernest Wu/2024 IEDM Tutorial Lecture

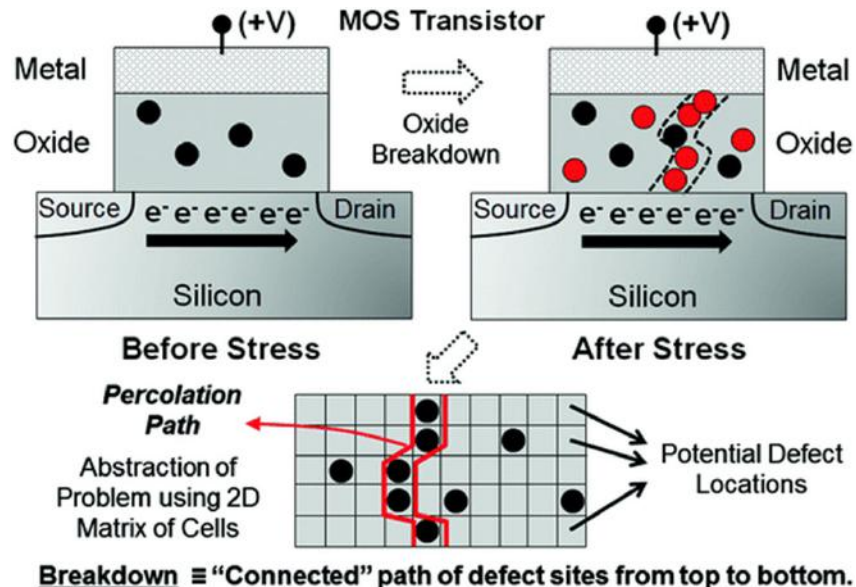


Miranda, IRPS 1998



Percolation Path:

- With the oxide is inserted to function as an insulator and the defect generation phenomenon is random in the oxide.
- When several defects are created and some of them form a cluster that connects the two ends of the network, we can interpret this to be a “breakdown” of the oxide as the fully connected path through the oxide is a favored region for high leakage currents to flow, resulting in electrical short and circuit failure.
- So, the defect generation and clustering process in dielectric results in a formation percolation path.



Gate dielectric Reliability and Failure analysis

Limit by convergence of HBD, SBD and DIBE

- **Soft breakdown (SBD)**
 - Gox has broken down
 - Device parameters such as gate leakage, I_{dsat} , transconductance etc. degrade
 - Device and circuit are still functioning
 - Two types of SBD
 - SBD without and with DBIE
- **Hard breakdown**
 - Catastrophic breakdown
 - Device totally malfunction

Stages of gate dielectrics degradation wear-out & progressive breakdown

3-stage process

1. Wear-out due to defect build up
 - TDDB
 - SILC
2. Gate oxide breakdown
 - Percolation path formation
 - Progressive breakdown
 - Formation of DBIE
3. Thermal runaway
 - DBIE shorts electrodes

Gate dielectric Reliability and Failure analysis

Soft Breakdown (SBD)

- Formation of **partial percolation paths** of traps/defects.
- Leakage increases gradually → **incremental conduction**.
- Common in **high-k/metal-gate stacks (HKMG)**.
- SBD may not immediately cause device failure, but **increases variability, noise, and subthreshold slope degradation**.

Hard Breakdown (HBD)

- **Catastrophic rupture**: creation of a permanent, low-resistance path.
- Usually occurs when SBD-induced filaments grow to become a full **conductive bridge**.
- Leads to:
 - Drain current runaway
 - Gate leakage $> 10^{-3}$ – 10^{-2} A/cm²
 - Device death (short to substrate/channel)

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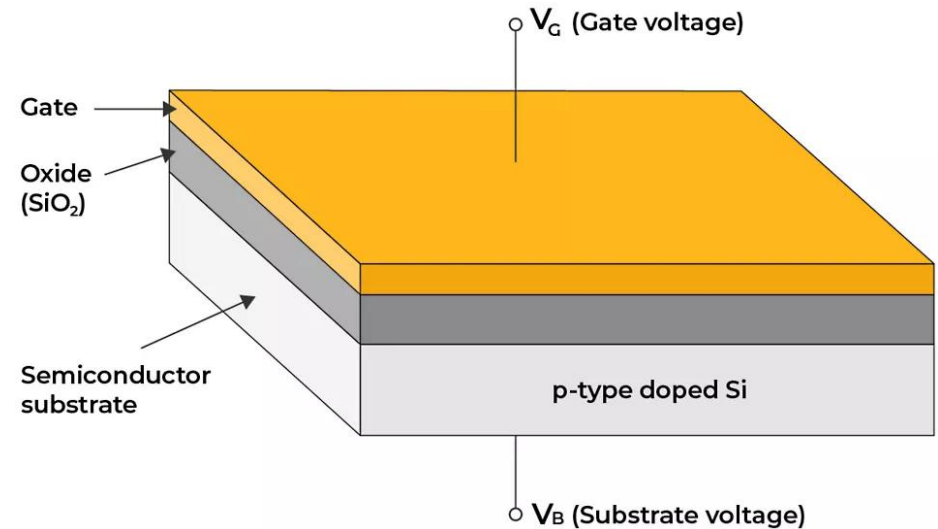
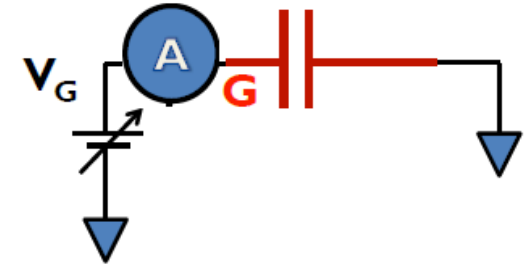
Gate dielectric Reliability and Failure analysis

Physics of oxide degradation and breakdown

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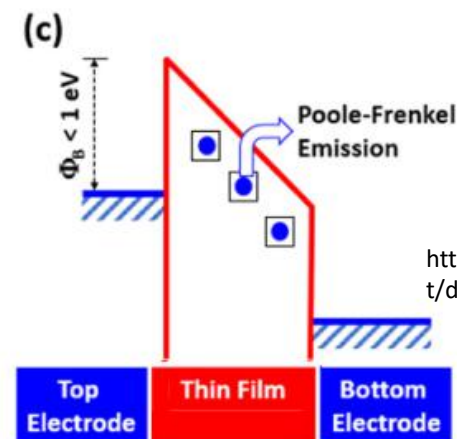
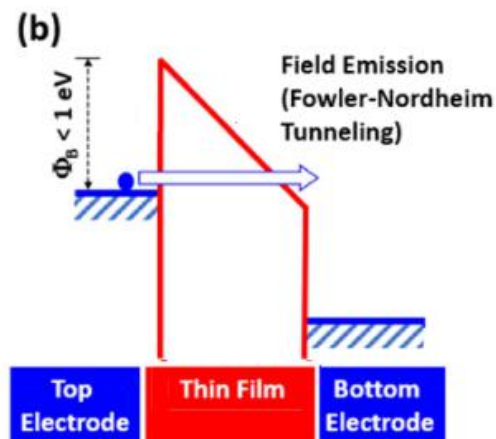
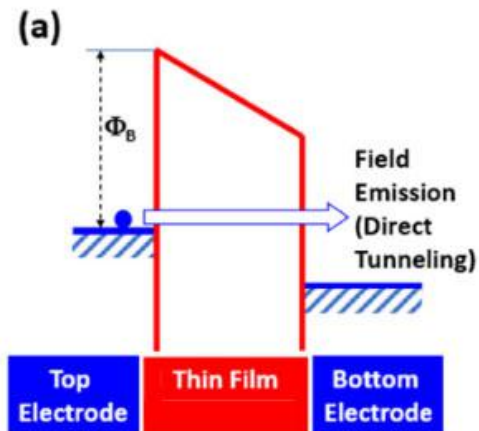
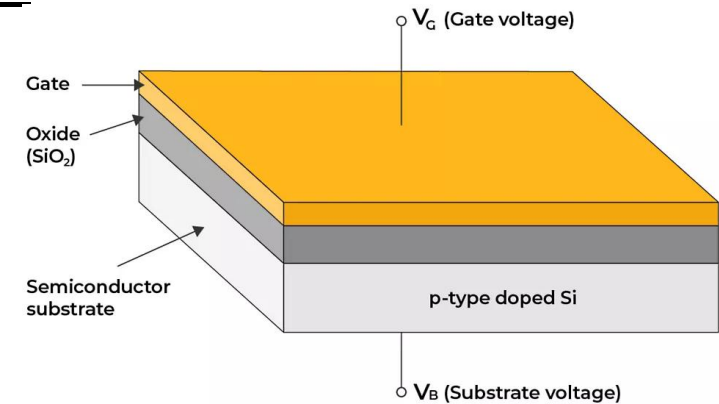
Gate dielectric leakage currents

- F-N Tunneling ($<5\text{nm } T_{ox}$)
- Direct Tunneling ($<2\text{nm } T_{ox}$)
- Trap-Assisted Tunneling (TAT)



Gate dielectric leakage currents

- Direct Tunneling ($< 2\text{nm } T_{ox}$)
- F-N Tunneling ($< 5\text{nm } T_{ox}$)
- Trap-Assisted Tunneling (TAT)



<https://www.degruyter.com/document/doi/10.1515/ntrev-2021-0047/html>

Gate dielectric Reliability and Failure analysis

Tunneling current in oxides

- Tunneling current will flow through oxide
 - High gate voltage
 - Thin oxide
- For thick oxide ($> 3\text{nm}$)
 - Fowler-Nordheim (FN) tunneling through triangular barrier
- For thin oxide ($< 3\text{nm}$)
 - Direct quantum tunneling (DT)
- Tunneling electrons will trigger processes in oxides

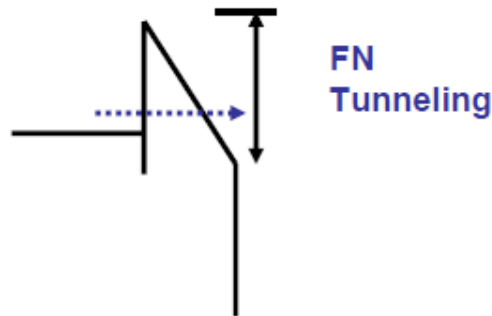
Gate dielectric Reliability and Failure analysis

Fowler-Nordheim (FN) Tunneling & Direct Tunneling

FN Tunneling

A triangular energy barrier can be observed.

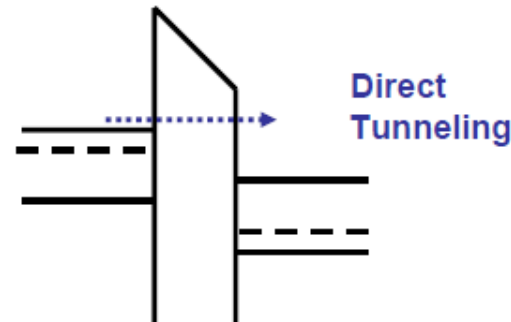
Tunneling is believed due to the “electrical thinning” of gate oxides.



Direct Tunneling

A trapezoidal energy barrier can be observed.

Direct tunneling starts to occur at $T_{ox} = 3-5$ nm and becomes more significant at $T_{ox} < 2$ nm.



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Gate dielectric Reliability and Failure analysis

FN Tunneling & Direct Tunneling

Low E-fields

Direct tunneling dominates over FN tunneling, i.e., no “electrical thinning” of gate oxides occurs.

Carrier's wave function penetration through the gate oxides is no longer negligible, leading to a higher tunneling current.

Intermediate E-fields

FN tunneling dominates, in which “electrical thinning” of gate oxides is strong enough to allow tunneling to occur.

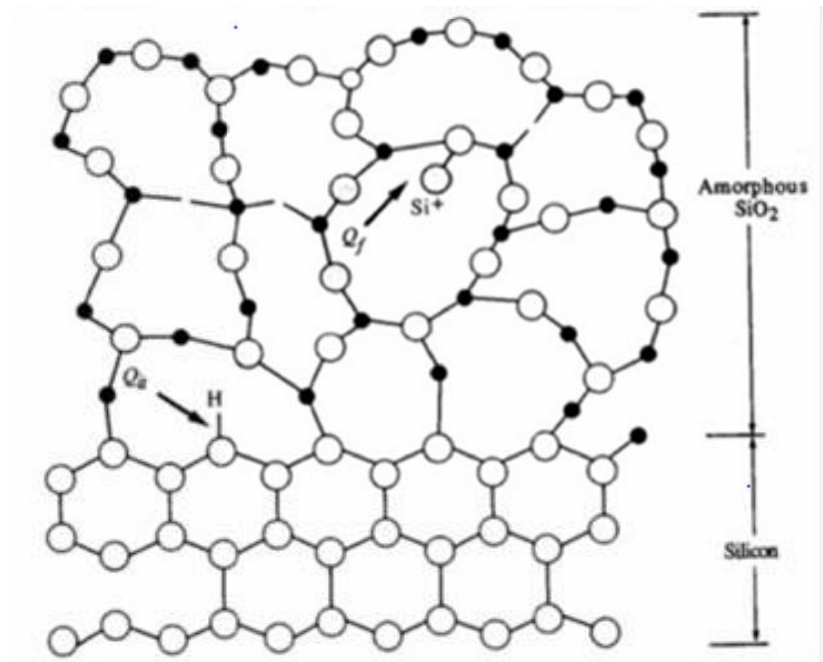
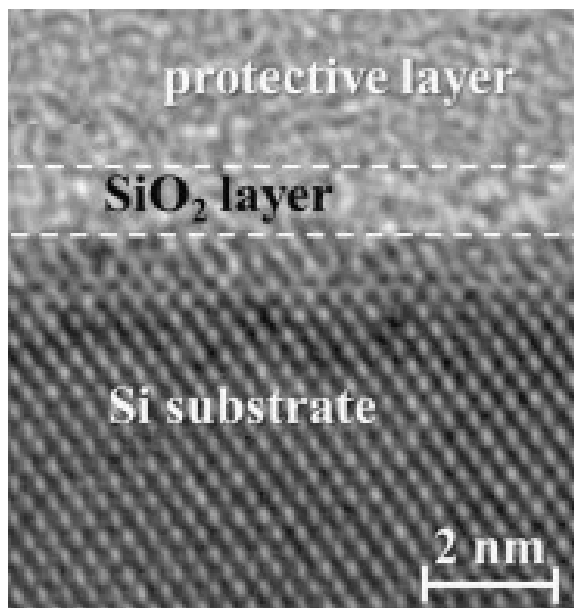
High E-fields

Breakdown occurs!!!

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Gate dielectric Reliability

SiO_2/Si



<https://www.sciencedirect.com/science/article/pii/S0927024815003207>

NE 343: Microfabrication and thin film technology

<http://ece.uwaterloo.ca/~bcui/>

Textbook: Silicon VLSI Technology by Plummer, Deal and Griffin

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Defect generation in gate dielectrics

- Hydrogen Release model
- Thermochemical model
- Anode hole injection
- Hot electron injection (HEI)

Defect generation mechanisms

- Higher voltages
 - Impact ionization and anode hole injection (AHI)
- Low energy process (1.2V - 5V)
 - Hydrogen release
- Damages within oxides build up, leading to a point where oxide breakdown

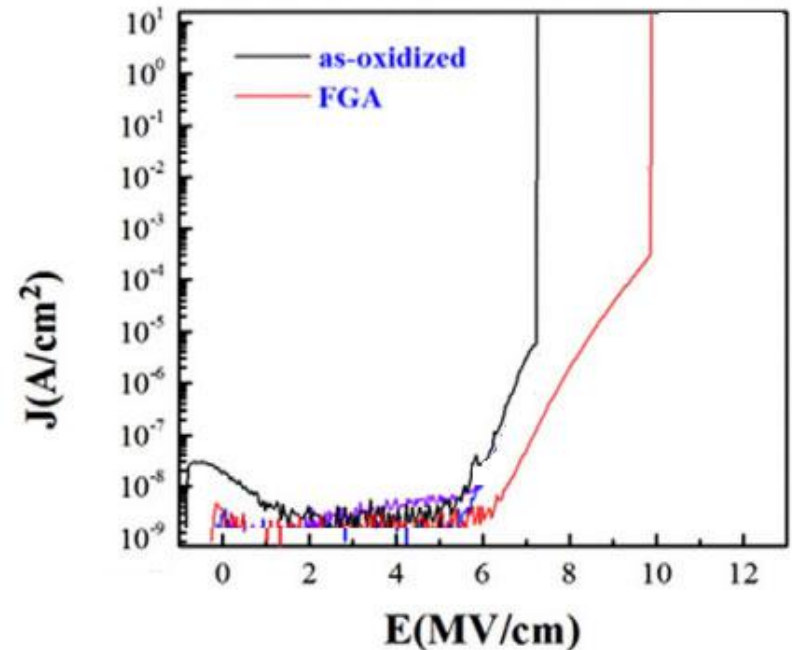
Gate dielectric Reliability

Forming Gas Annealing (FGA): is a post-deposition annealing process used to improve the electrical properties of dielectric materials in semiconductor devices.

The gas mixture typically consists of **H₂ and N₂**, and the process is carried out at moderate temperatures (~300-500°C).

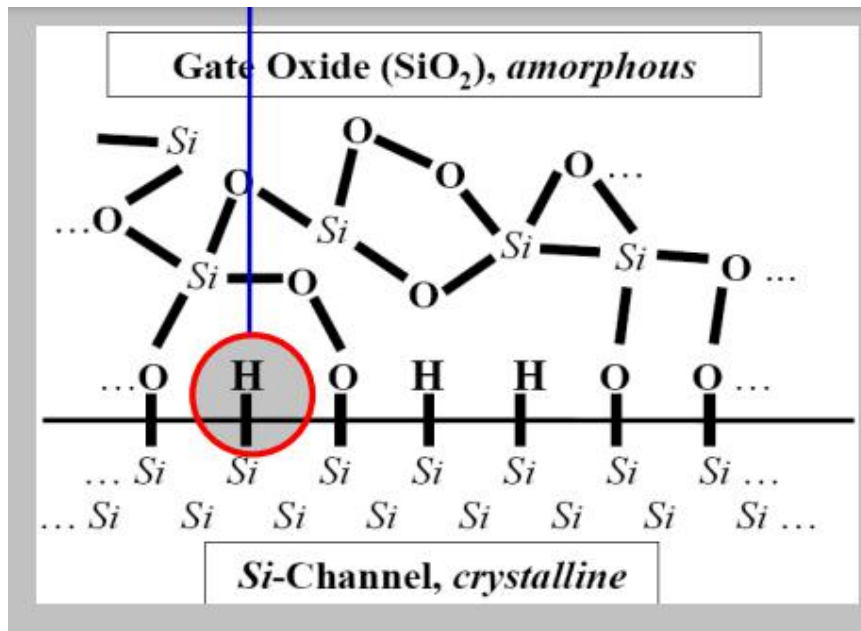
Reduction of Interface Traps (Defect Passivation)

- Hydrogen and Nitrogen atoms diffuse into the dielectric and interface, passivating dangling bonds (e.g., Si-O, Si-N at the Si/SiO₂ interface).
- This reduces interface trap density (D_{it}), improving carrier mobility and subthreshold swing in MOSFETs.



Main Defect generation mechanism in gate dielectrics

1. Hydrogen Release model

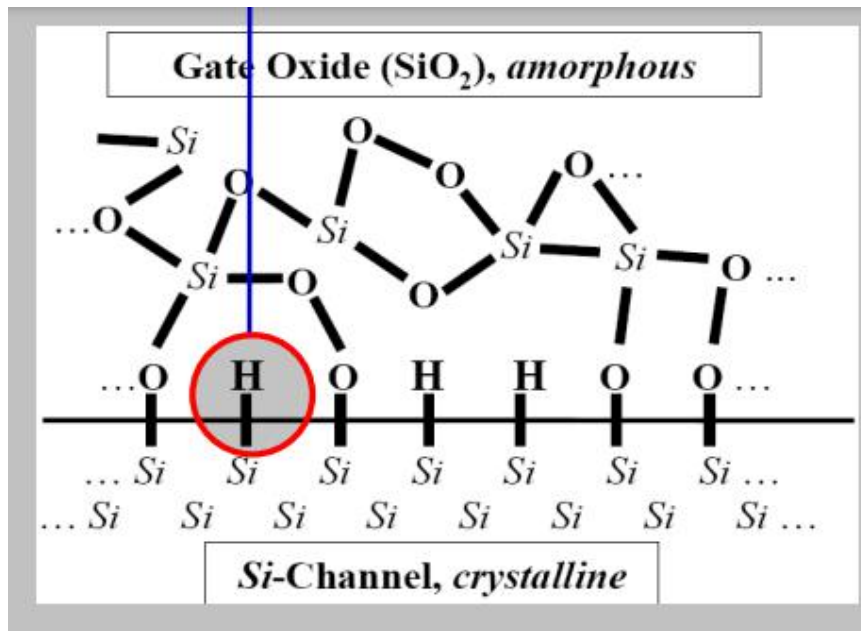


The hydrogen release model provides insights into the mechanisms behind dielectric degradation due to Si-H bond breakage

- The release of hydrogen can lead to void formation at the Silicon-dielectric interface
- Hydrogen released contributes to the degradation process of the dielectric by promoting chemical reactions.
- This phenomenon is particularly relevant in high-voltage applications, more relevant in NBTI process (in pMOSFET).

Main Defect generation mechanism in gate dielectrics

1. Hydrogen Release model

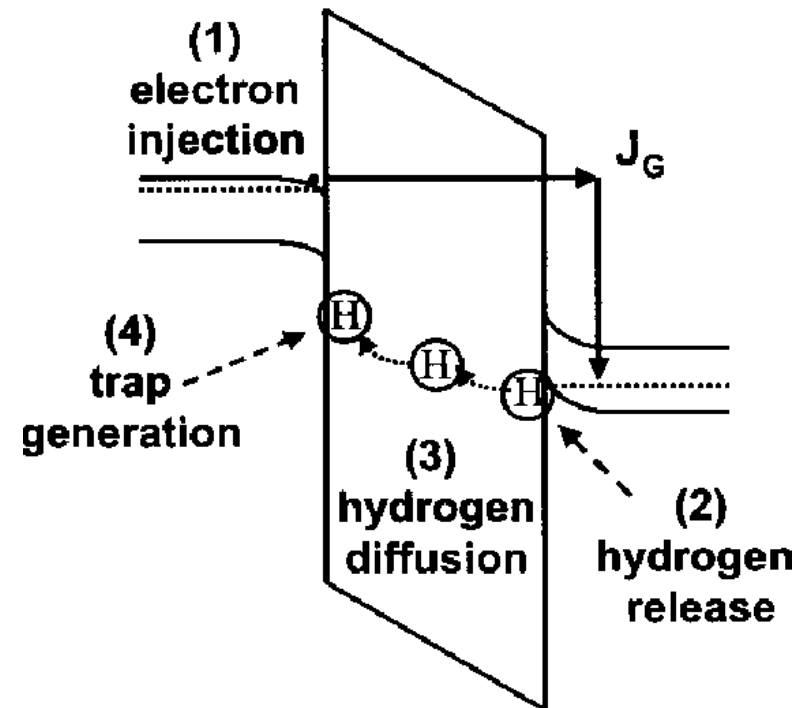
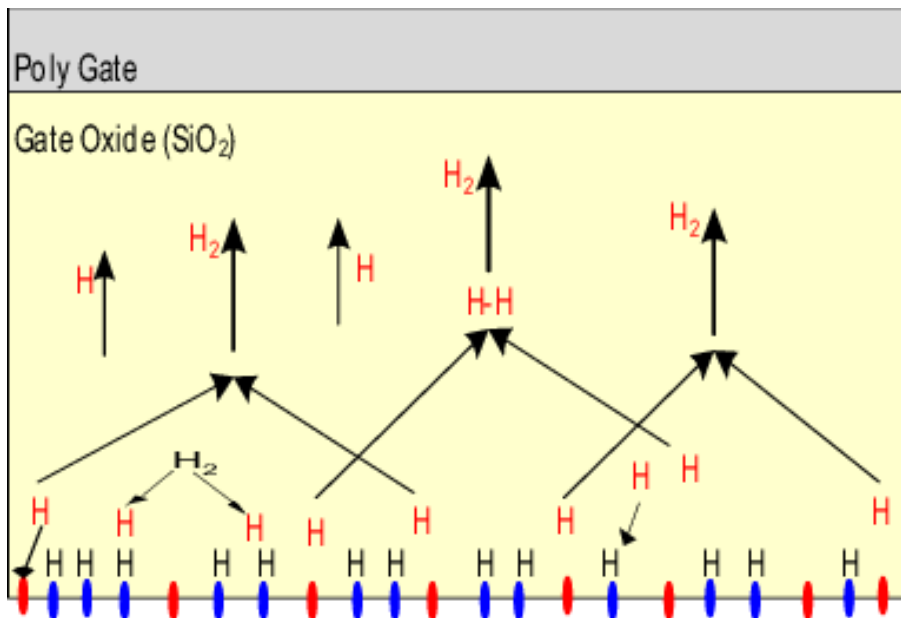


Hydrogen Release Model (HRM) is one of the most widely accepted physical models explaining **Time-Dependent Dielectric Breakdown (TDDB)** and **Bias Temperature Instability (BTI)** in gate dielectrics.

- Hydrogen atoms (or H-related species) stored in the dielectric or at the interface are released under high electric field/temperature, break Si–O or Si–H bonds, and generate defects that lead to degradation and eventually breakdown.

Defect generation mechanism in gate dielectrics

1. Hydrogen Release model

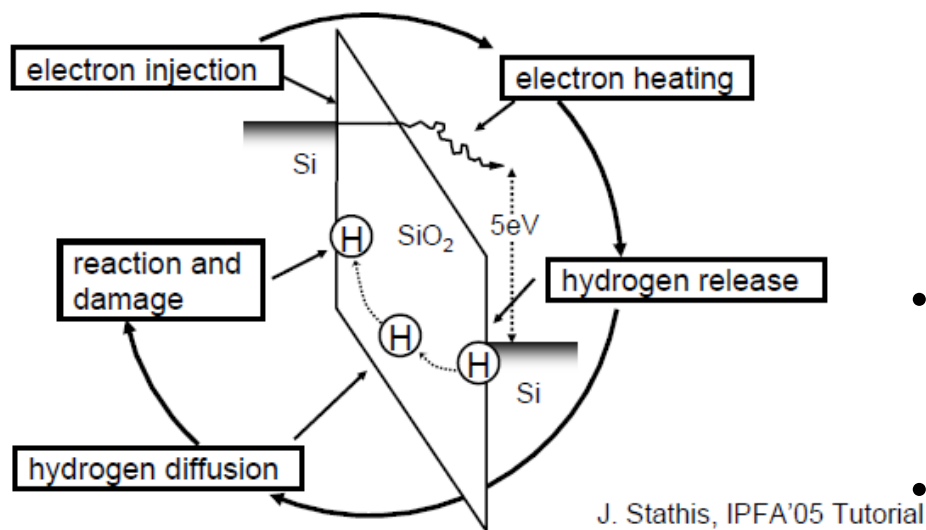


S. Khan, et al., Temperature dependence of NBTI induced delay, IOLTS 2010), 5-7 July, 2010

DOI: [10.1109/IOLTS.2010.5560238](https://doi.org/10.1109/IOLTS.2010.5560238)

Defect generation in gate dielectrics

1. Hydrogen mechanism of oxide degradation



J. Stathis, IPFA'05 Tutorial

- Hydrogen Release Model explains that electric-field/thermal stress causes Si–H and O–H bonds to break, releasing hydrogen that diffuses through the dielectric, creating defect traps that ultimately form a conductive path and cause dielectric breakdown.
- Eventually, enough defects form to create a **percolation path** from gate to substrate.
- Hydrogen release accelerates the BD process.

E. Wu PRL, 2005

K.L. Pey, Course material - Gate oxide dielectric failure mechanisms and reliability-NTU

Defect generation in gate dielectrics: Thermochemical model

The Thermochemical Model states:

Dielectric breakdown occurs because high electric field exerts a force that *weakens and breaks chemical bonds* in the dielectric. The rate of bond-breaking increases exponentially with temperature and field.

It is *chemically activated bond rupture*, not purely electronic.

Thermochemical model

- Emphasizes on **thermal** driven degradation mechanisms in the dielectrics
- Defect generation is **field driven**

Defect generation in gate dielectrics

- The thermochemical model of dielectric degradation focuses on the interactions between thermal energy and chemical processes that lead to the deterioration of dielectric materials.
- This model is particularly relevant in understanding how heat and electrical stress contribute to the breakdown of dielectrics over time.

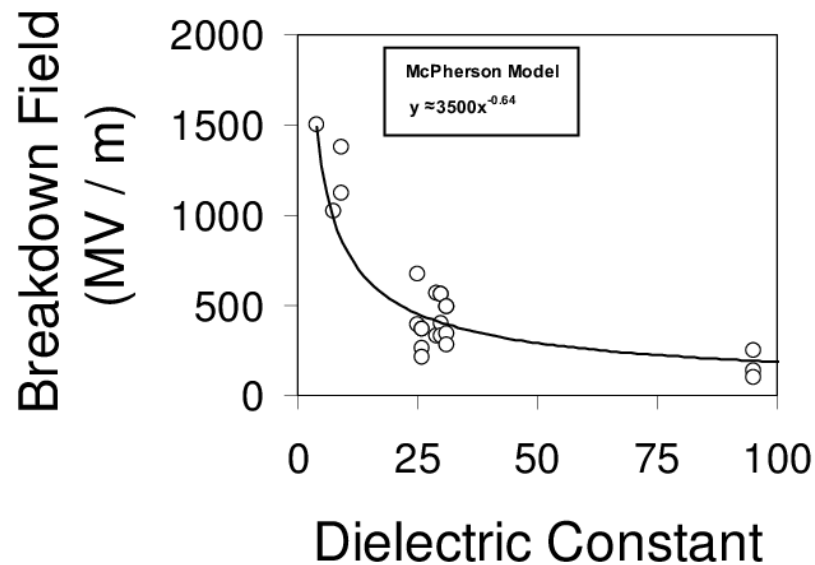
Each Si–O, Hf–O, Si–N, or O–H bond has a bond energy:

$$E_b$$

A high electric field **adds mechanical force** to the bond, reducing the required energy to break it.

Defect generation in gate dielectrics

The thermochemical reaction model explains dielectric degradation as bond-breaking due to electric-field-assisted lowering of chemical bond energies, with temperature accelerating the defect generation that leads to breakdown.



https://www.researchgate.net/publication/216619328_Prospects_for_the_ultimate_energy_density_of_oxide-based_capacitor_anodes

Defect generation in gate dielectrics

When a high electric field is applied across the gate dielectric, **electrons injected from the cathode** (the negatively biased electrode) enter the oxide and accelerate toward the **anode** (the positively biased electrode).

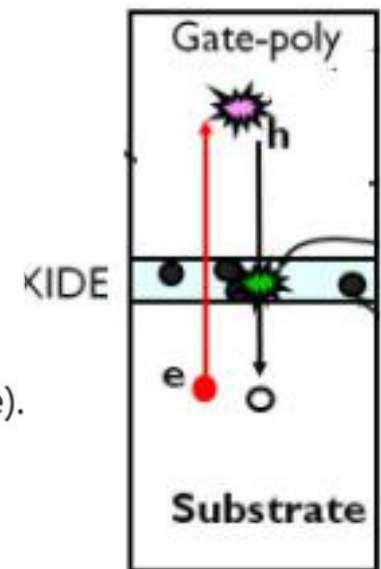
When these electrons reach the anode, they have gained enough kinetic energy to:

- ✓ Impact-ionize the anode material

This creates **electron-hole pairs (EHPs)** inside the anode (e.g., poly-Si gate or metal gate).

- ✓ Holes generated at the anode are injected back into the oxide

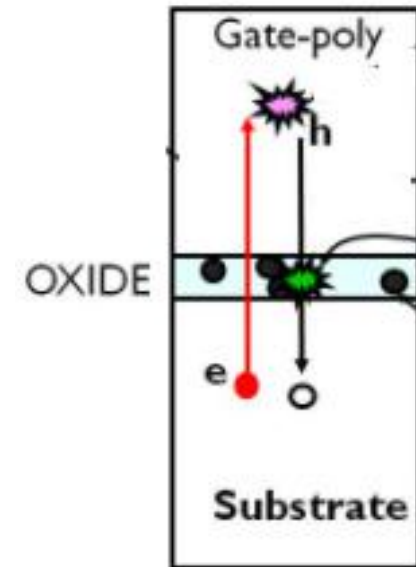
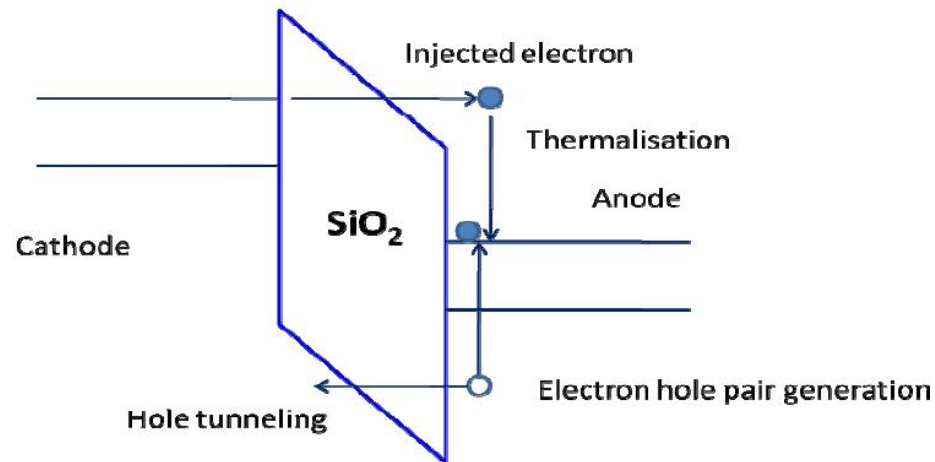
These **energetic holes** enter the dielectric and cause degradation by breaking chemical bonds.



Defect generation mechanism in gate dielectrics

3. Anode hole injection model

- Hole trapping at cathode (injecting surface)
- Electron injection may cause impact ionization at SiO_2 layer
- Localized high field/current density region at the weak spot

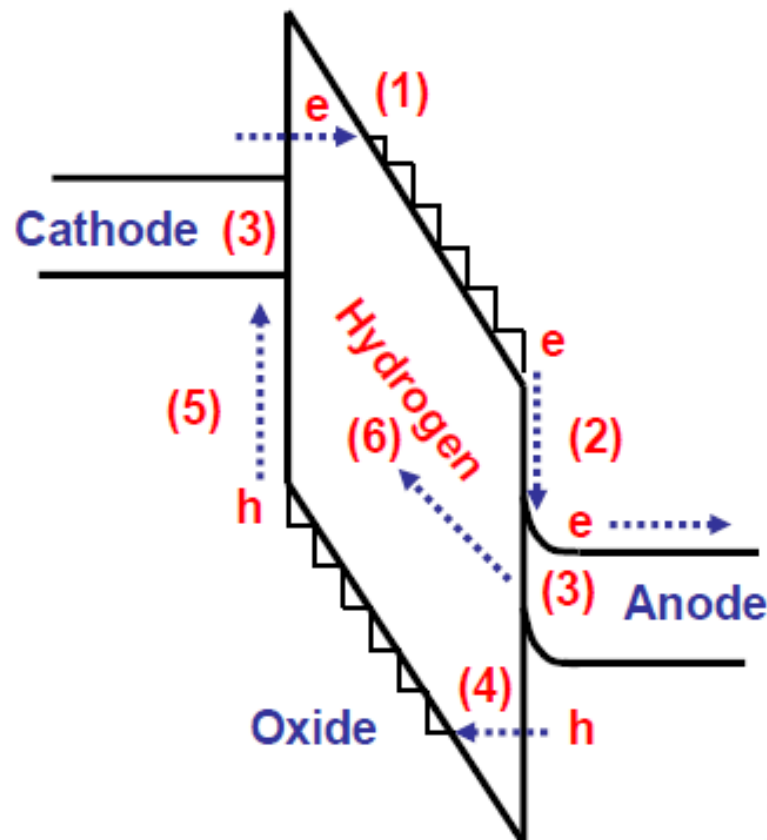


M. A. Alam, Reliability Physics of Nanotransistors, Purdue University;
www.nanohub.org

Dielectric Degradation Mechanism

Steps for the dielectric degradation mechanism that was proposed by DiMaria *et al.*

- (1) Hot electron injection
- (2) Energy released by hot electrons
- (3) Bond breaking – trap generation
- (4) Hot hole generation and injection
- (5) Energy released by hot holes
- (6) Hydrogen release – trap generation



K.L. Pey, Course material - Gate oxide dielectric failure mechanisms and reliability-NTU

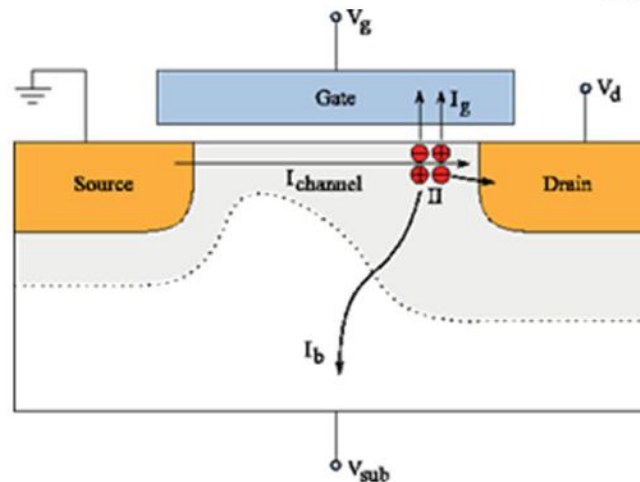
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Hot carrier effect: Defect generation mechanism

In nanoscale MOSFETs, strong electric fields near the drain accelerate carriers into “hot carriers” that impact-ionize the lattice and inject into the gate dielectric, breaking bonds and generating traps that degrade MOSFET performance and reliability over time.

Hot carriers are **electrons or holes** that **gain very high kinetic energy** from the electric field in the transistor channel. Because of their energy, they can:



<https://www.iue.tuwien.ac.at/phd/entner/node21.html>

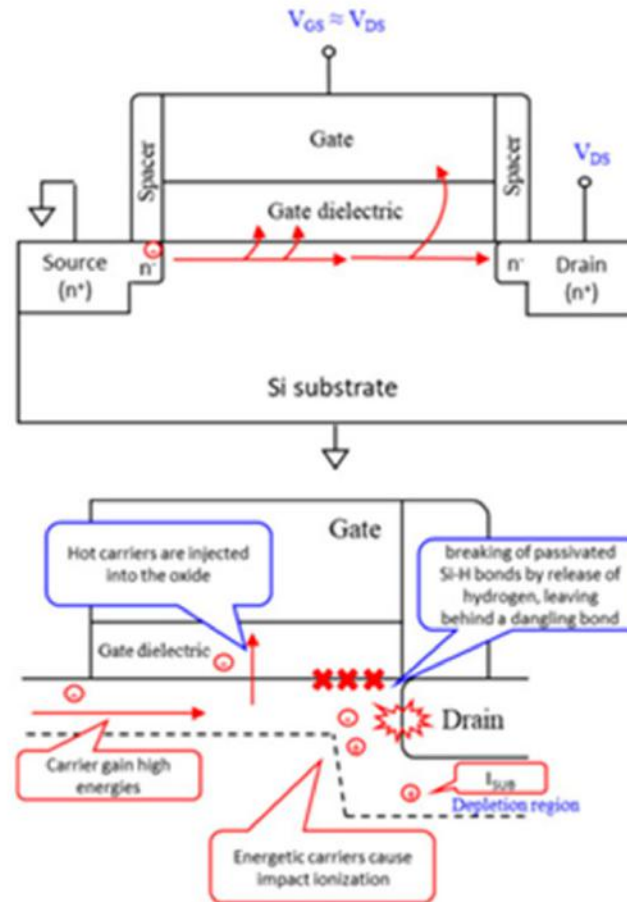
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Hot carrier effect: Defect generation mechanism

Hot carriers are electrons or holes that gain very high kinetic energy from the electric field in the transistor channel. Because of their energy, they can:

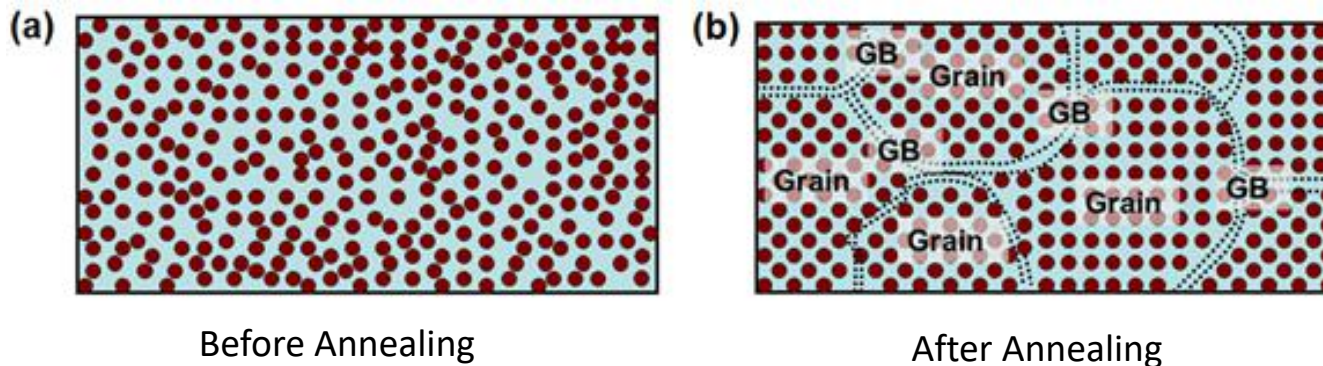
- Damage the Si/SiO₂ or high-k gate stack
- Generate interface traps
- Increase threshold voltage
- Reduce mobility and transconductance
- Cause long-term device degradation



<https://www.mdpi.com/2079-9292/9/12/2095>

Issues with High- κ Gate Dielectrics

- Defective bulk, formation of SiO_2 interfacial layer
- Polycrystalline microstructure
 - Grain boundary
- Requires compatible gate electrode material
- Compatibility with the existing Si technology



HfO₂ dielectrics -Grain Boundary Effect

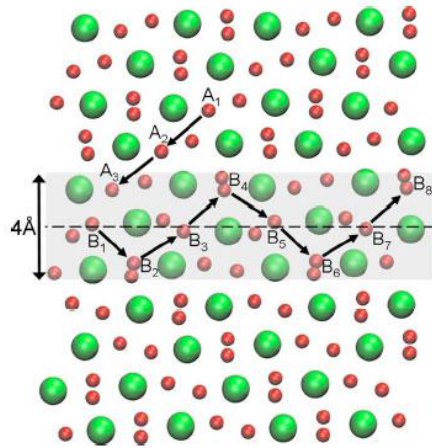


FIG. 1. (Color online) Atomic structure near the (101) twin boundary in m-HfO₂, Hf ions (large spheres) and O ions (small spheres). The 4 Å wide region near the GB where vacancy segregation is most favorable is indicated. Paths for vacancy diffusion toward (A) and along (B) the GB are also shown (see text for description).

K. McKenna and A. Shluger, APL, 95, 222111 (2009).

- Vacancy segregation to the GB is preferred
- GB paths have lower diffusion barrier compared to grain

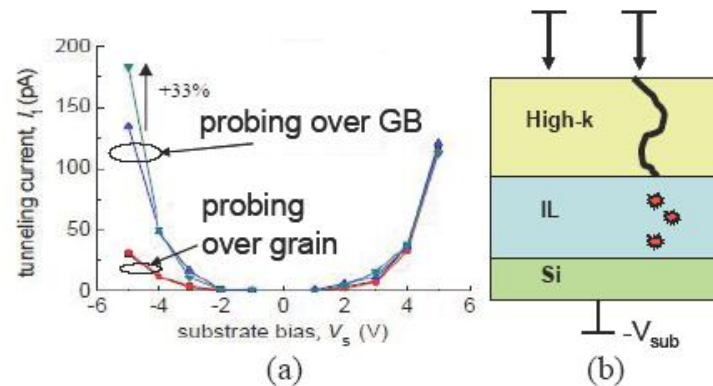
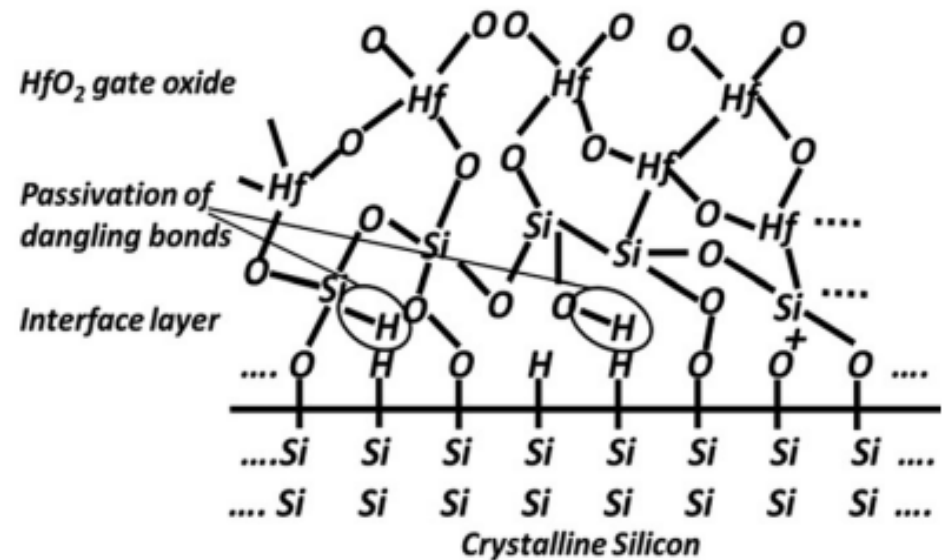
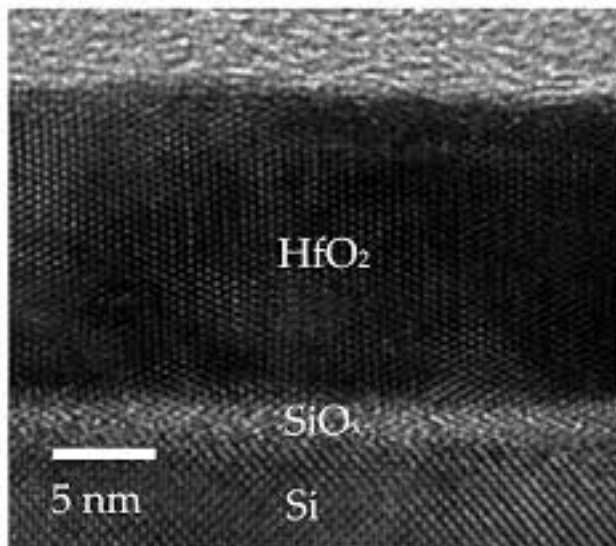


Fig. 8 (a) Initial and post-stress tunneling currents measured over the grain and GB (b) Schematic of the STM probing over the grain and GB and stress-induced defects generated in IL.

G. Bersuker, et al., IRPS , 373 (2010).

Gate dielectric Reliability and Failure analysis

HfO₂/SiO_x/Si

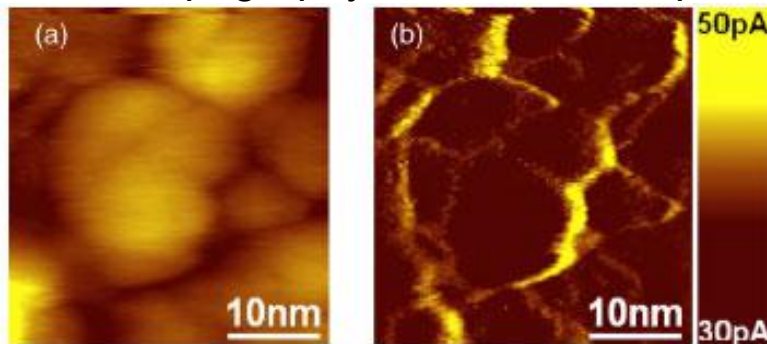


S. Maurya, et al., APPLIED PHYSICS LETTERS 105, 071605 (2014)

X.-Y. Chang, et al., Appl. Sci. 2017, 7, 1244; doi:10.3390/app7121244

HfO₂ dielectrics - Grain Boundary Effect

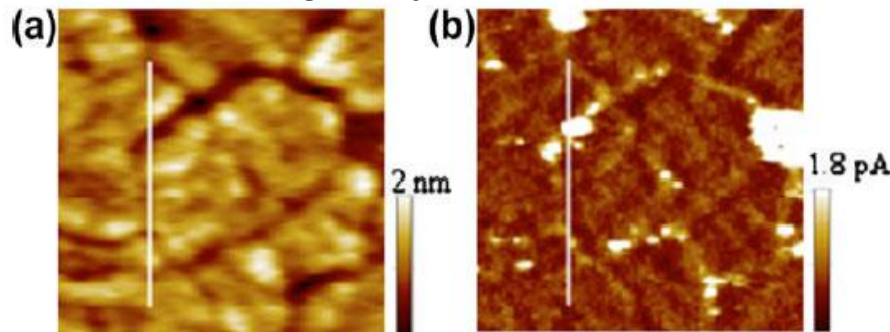
STM topography and current map



- Higher leakage current GB sites.

K. S. Yew, et al., EDL, 33, 741 (2012).

C-AFM topography and current map



- Bright leakage spots corresponding to GBs

K. McKenna, et al., Microele. Eng., 88, 1272 (2011).

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Physical analysis of Soft BD and Hard BD in dielectrics

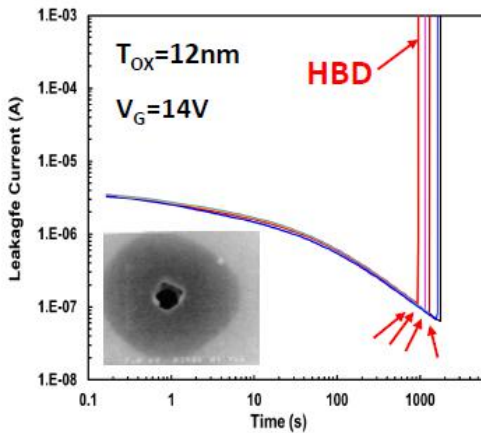
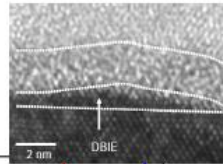
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Gate dielectric Breakdown

- BD is a loss of insulating property, a stochastic event with a discontinuity in I-t trace

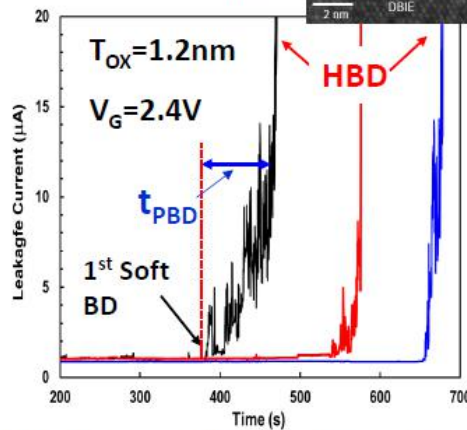
- BD is hard and instantaneous

- BD is soft with a delay (t_{PBD})



Pin-hole: Vollertsen MR 2013

Tutorial 4



DBIE: Dielectric BD induced epitaxy,
Pey et al. IRPS 2006

Ernest Wu/2024 IEDM Tutorial Lecture

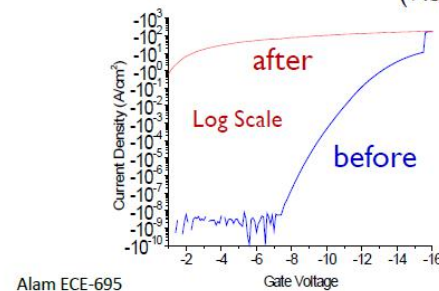
Hard Dielectric Breakdown



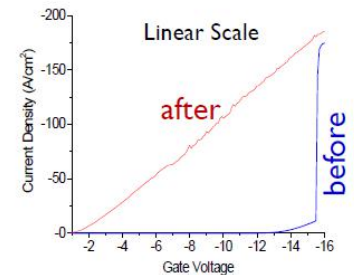
The poly-Si electrode has melted away over 20mm²

D. R. Volters and J. F. Verwey, "Breakdown and Wearout Phenomena in SiO₂ Films," Chap. 6 p.329, 1986.

(9.5nm oxide)

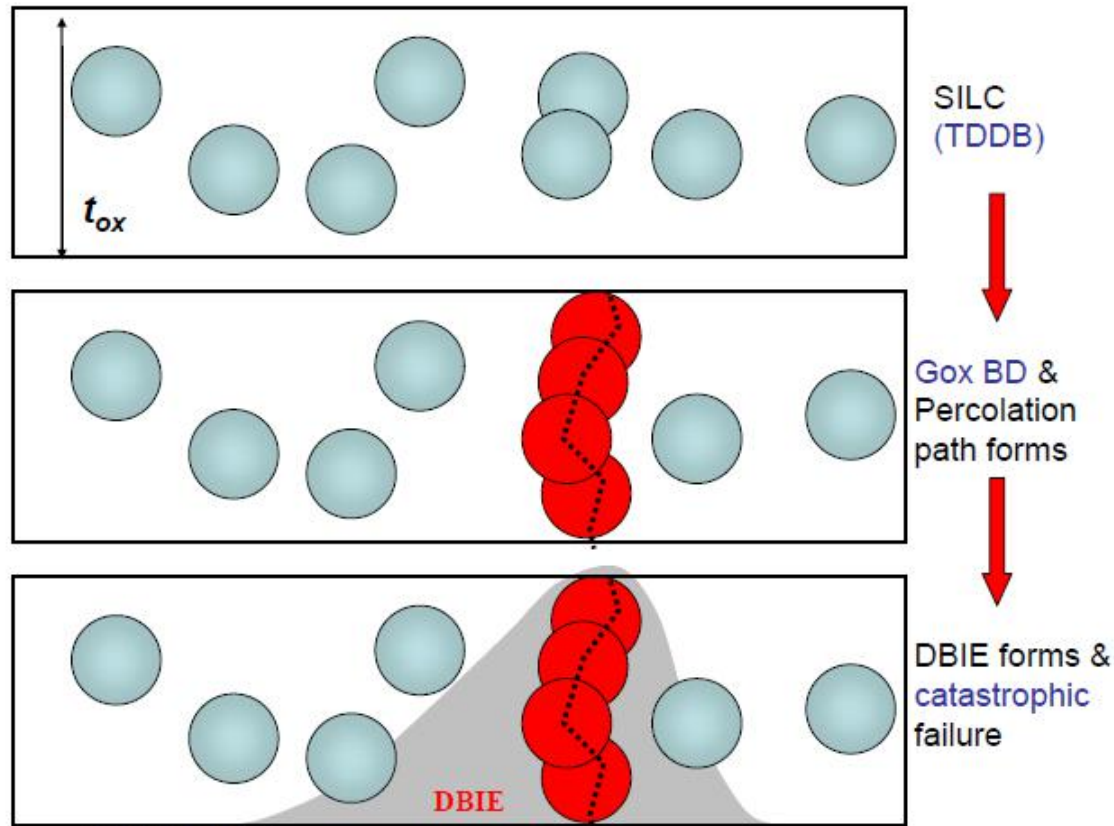


Alam ECE-695



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Gate dielectric Reliability and Failure analysis



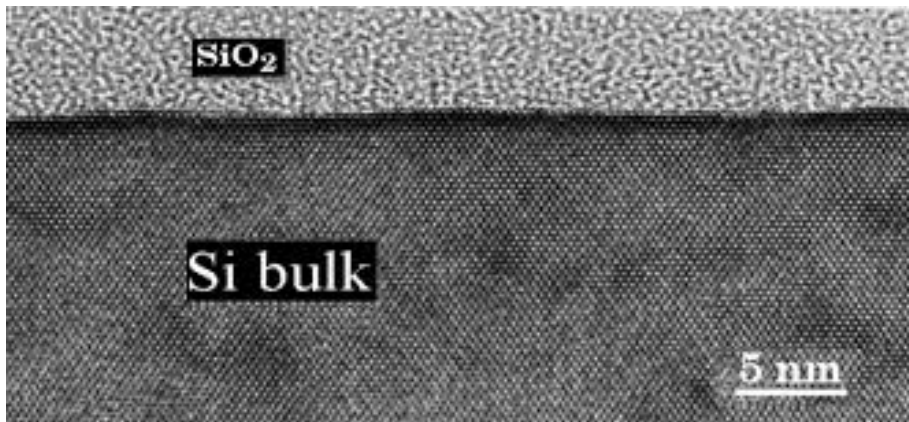
© Copyright by PKL

K.L. Pey, Course material - Gate oxide dielectric failure mechanisms and reliability-NTU

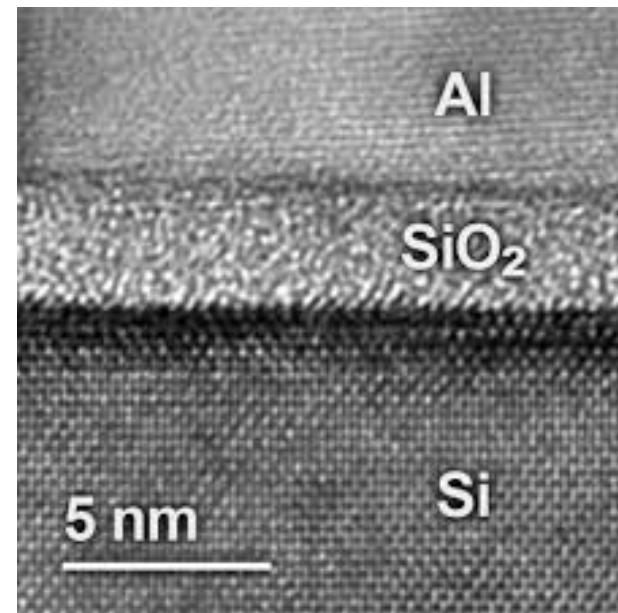
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Physical analysis of Soft BD and Hard BD in SiO₂ dielectrics

Digital BD initial phase of BD:

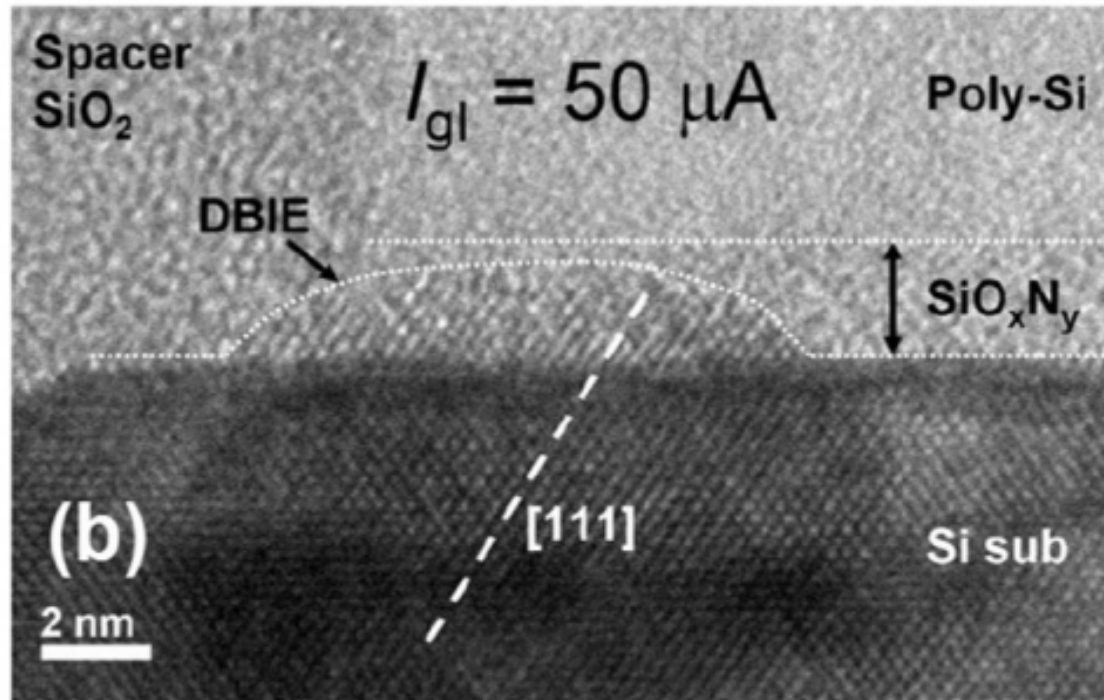


- No visible microstructural physical damage
- Percolation path formation
- Local electrical and chemical property changes



Adv. Nat. Sci.: Nanosci. Nanotechnol. 1 (2010) 043004 (11pp)

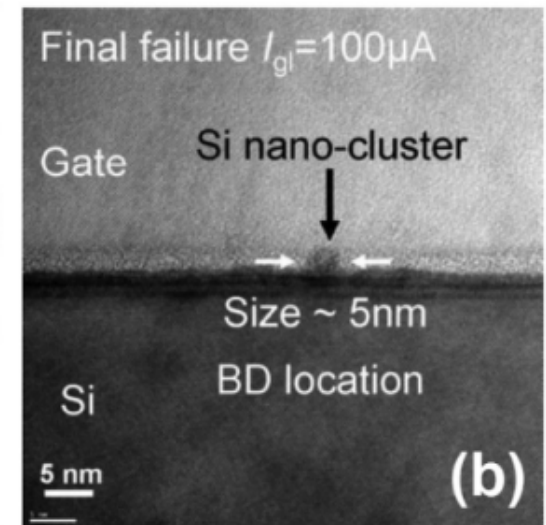
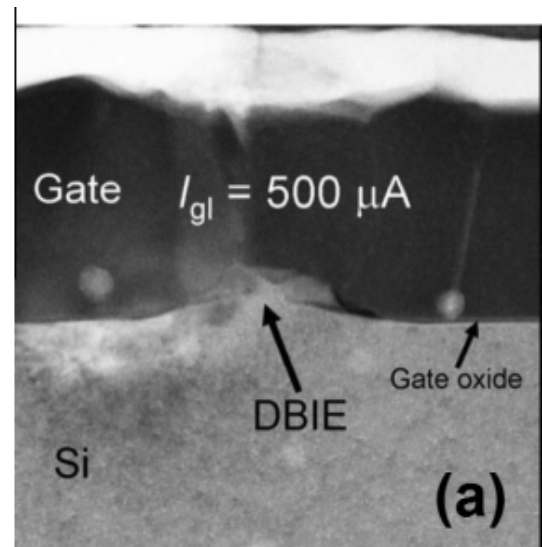
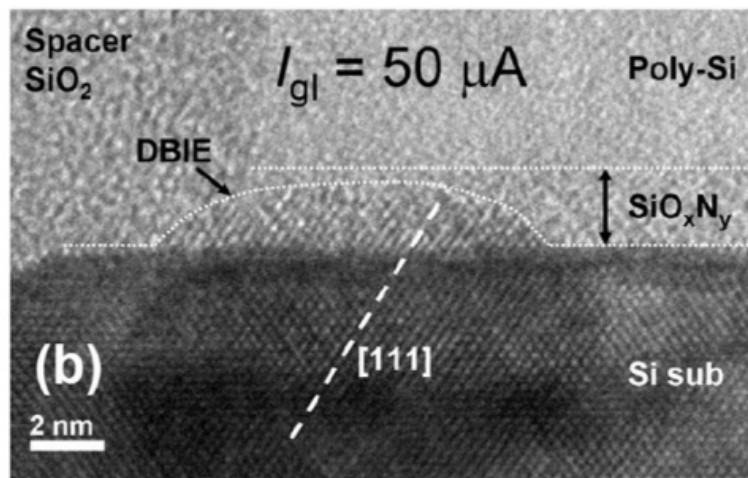
Physical analysis of Hard BD in SiO₂ dielectrics



- Visible damage and microstructural changes
- BD path formation

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Physical analysis of Hard BD in SiO₂ dielectrics



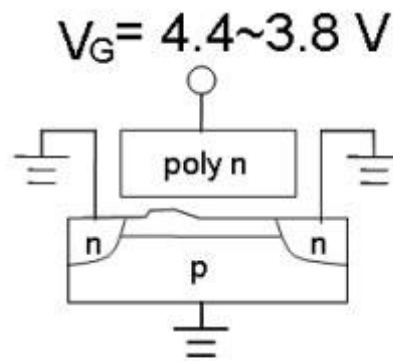
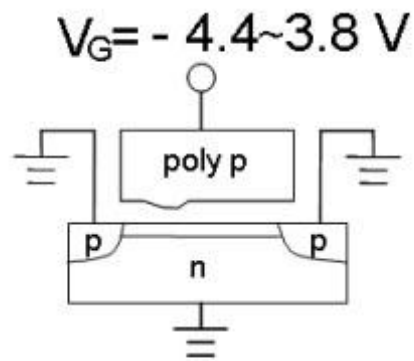
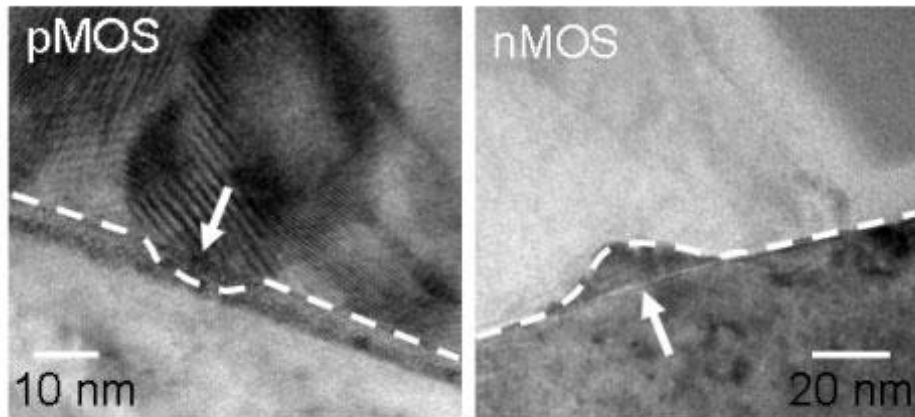
Microelectronics Reliability 54 (2014) 847–860

- Visible damage and microstructural changes
- BD path formation

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Physical analysis of Hard BD in SiO₂ dielectrics



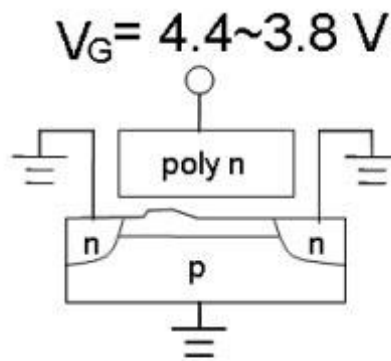
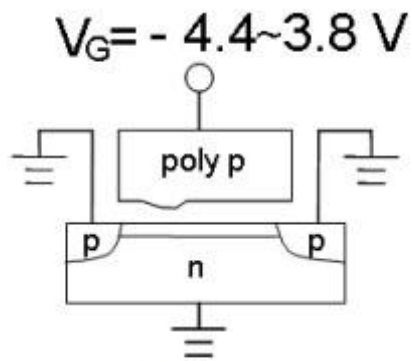
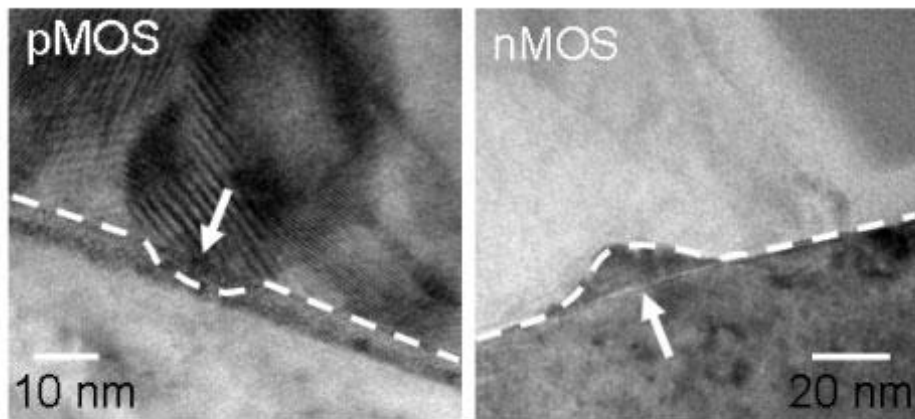
TEM micrograph for a PMOS and NMOS device showing the DBIE epitaxial defect nucleating from the cathode side and pinging into the oxide layer

S3Wflmf{fs1jyfQlHtskjwshj?Nylwfyi Hmhzyx1KNH 5>3Uwhjjinslx tkymj 755>

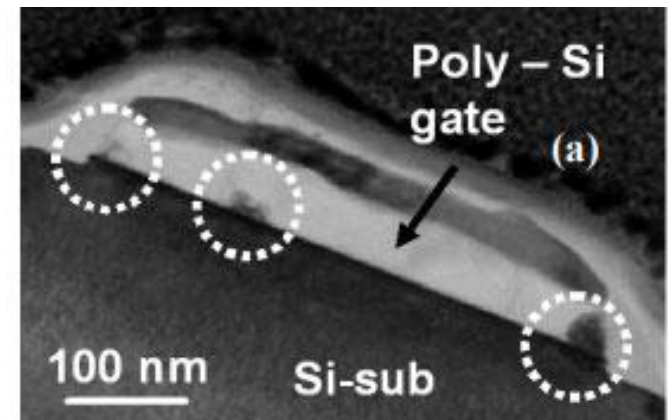
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Physical analysis of Hard BD in SiO₂ dielectrics



TEM micrograph for a PMOS and NMOS device showing the DBIE epitaxial defect nucleating from the cathode side and pinging into the oxide layer



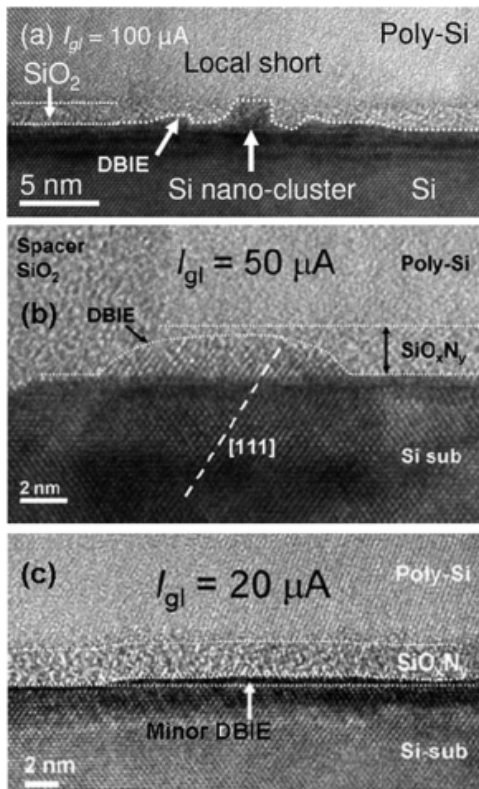
TEM cross-section of a large area NMOS device showing the three different breakdown locations

S3Wflmf{fs1jyfQlHtskjwshj?Nylwfyi Hmhzn1KNH 5>3Uwhjjinslx tkymj 755>

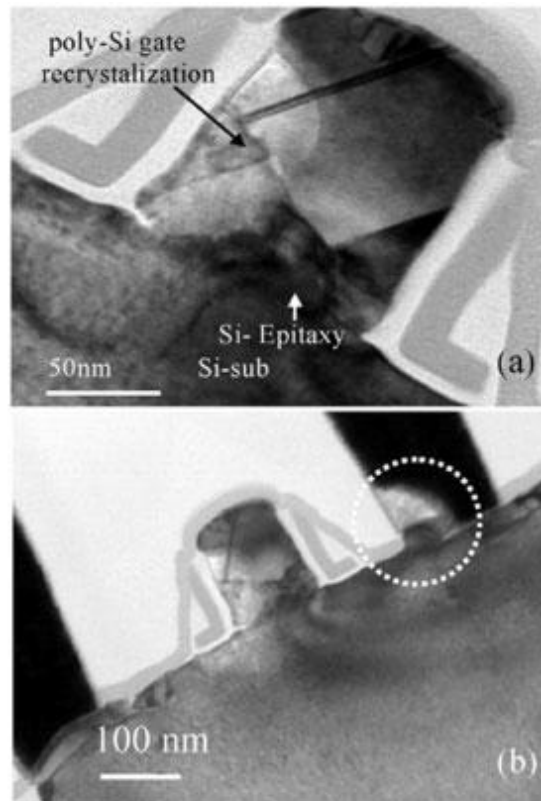
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Physical analysis of Hard BD in SiO₂ dielectrics

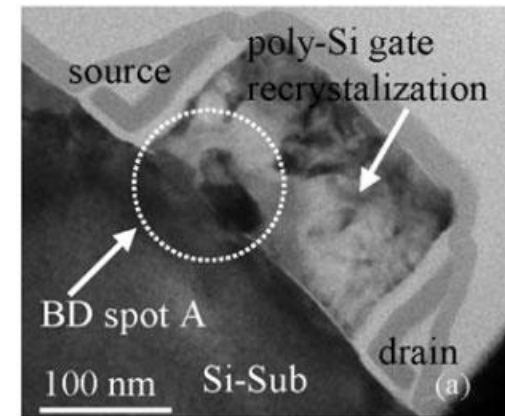


J. Appl. Phys. 114, 094504 (2013)



TEM micrographs of an HBD nMOSFET (i.e., HBD1 at 3.0 V)
 (a) DBIE and poly-Si gate electrode recrystallization. (b) Partial tungsten plug burnt-out (indicated by dotted circle).

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY, VOL. 5, NO. 2, JUNE 2005

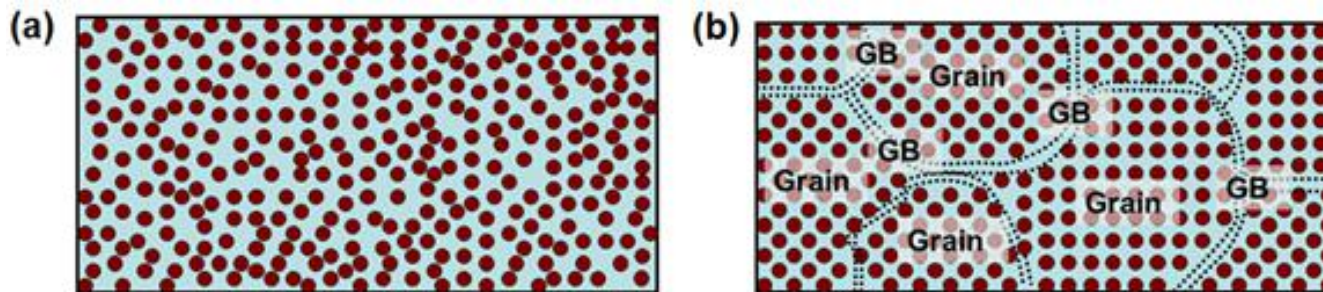


TEM micrographs of an HBD nMOSFET

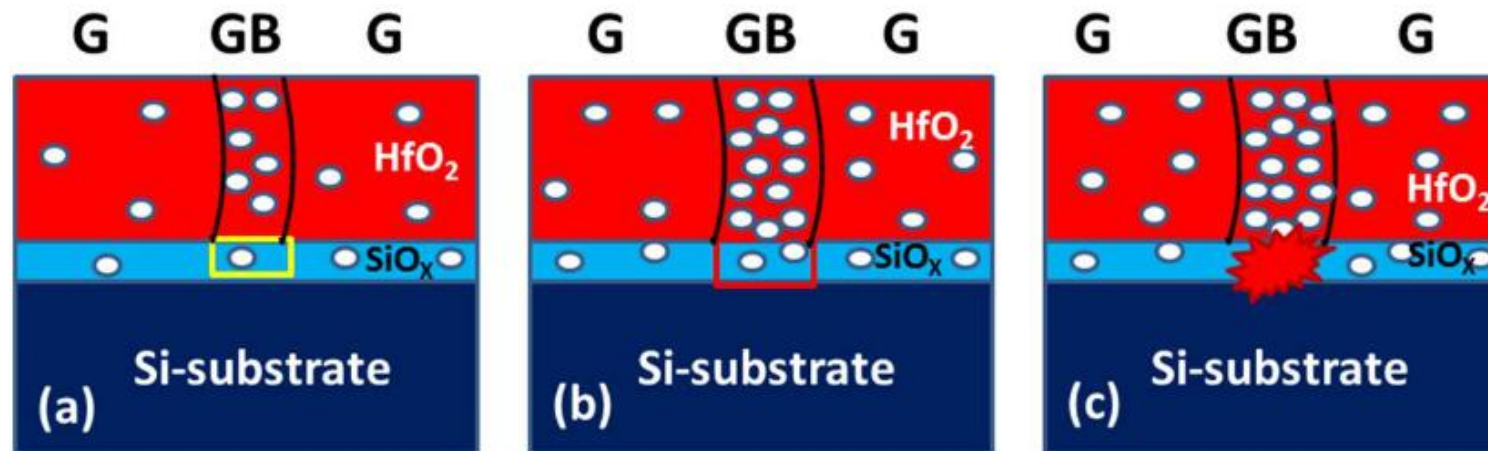
IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY, VOL. 5, NO. 2, JUNE 2005

High- κ Gate Dielectrics

- Defective bulk, formation of SiO_2 interfacial layer
- Polycrystalline microstructure
 - Grain boundary
- Requires compatible gate electrode material
- Compatibility with the existing Si technology

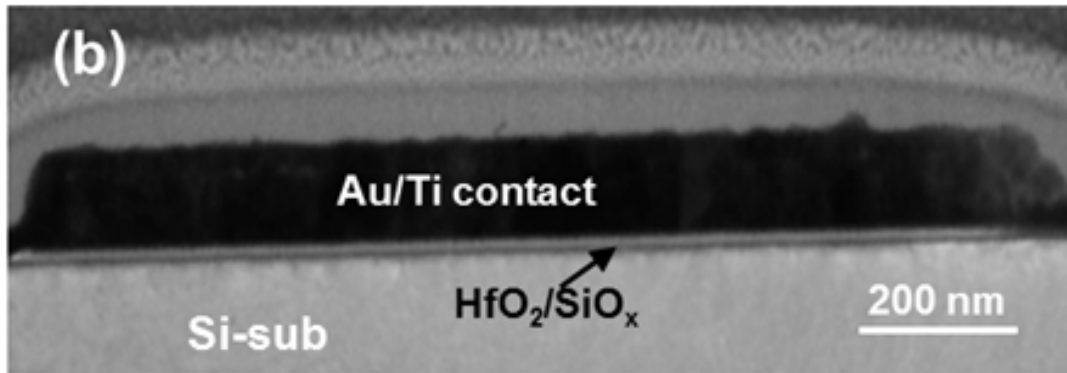


BD Sequence of HK/IL stacks



- Schematics of Proposed Degradation and BD sequence of individual layers in HK/IL dielectric stacks

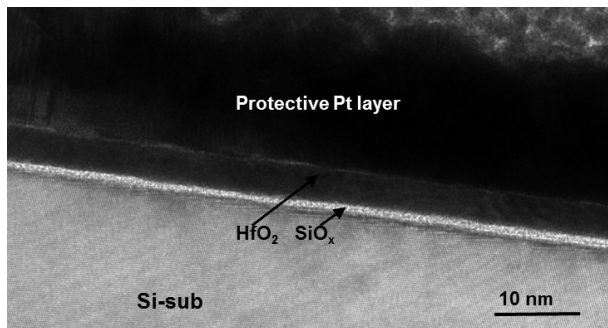
BD Analysis of HK/IL stacks



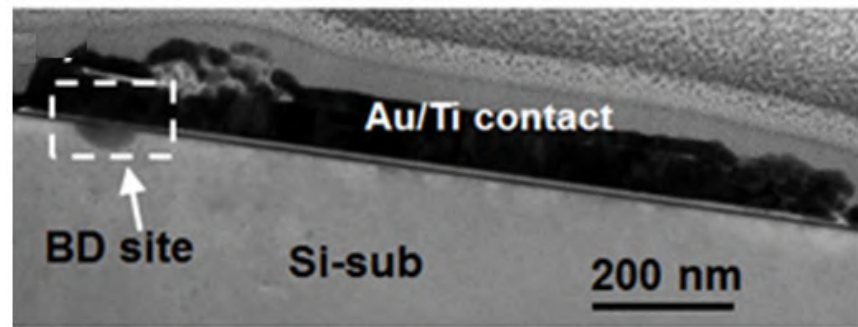
- No visible microstructural physical damage
- Percolation path formation
- Local electrical and chemical property changes

Increased BD Hardness in HK/IL stacks

Blanket region (B2)



Gate electrode region (G2)



Blanket:

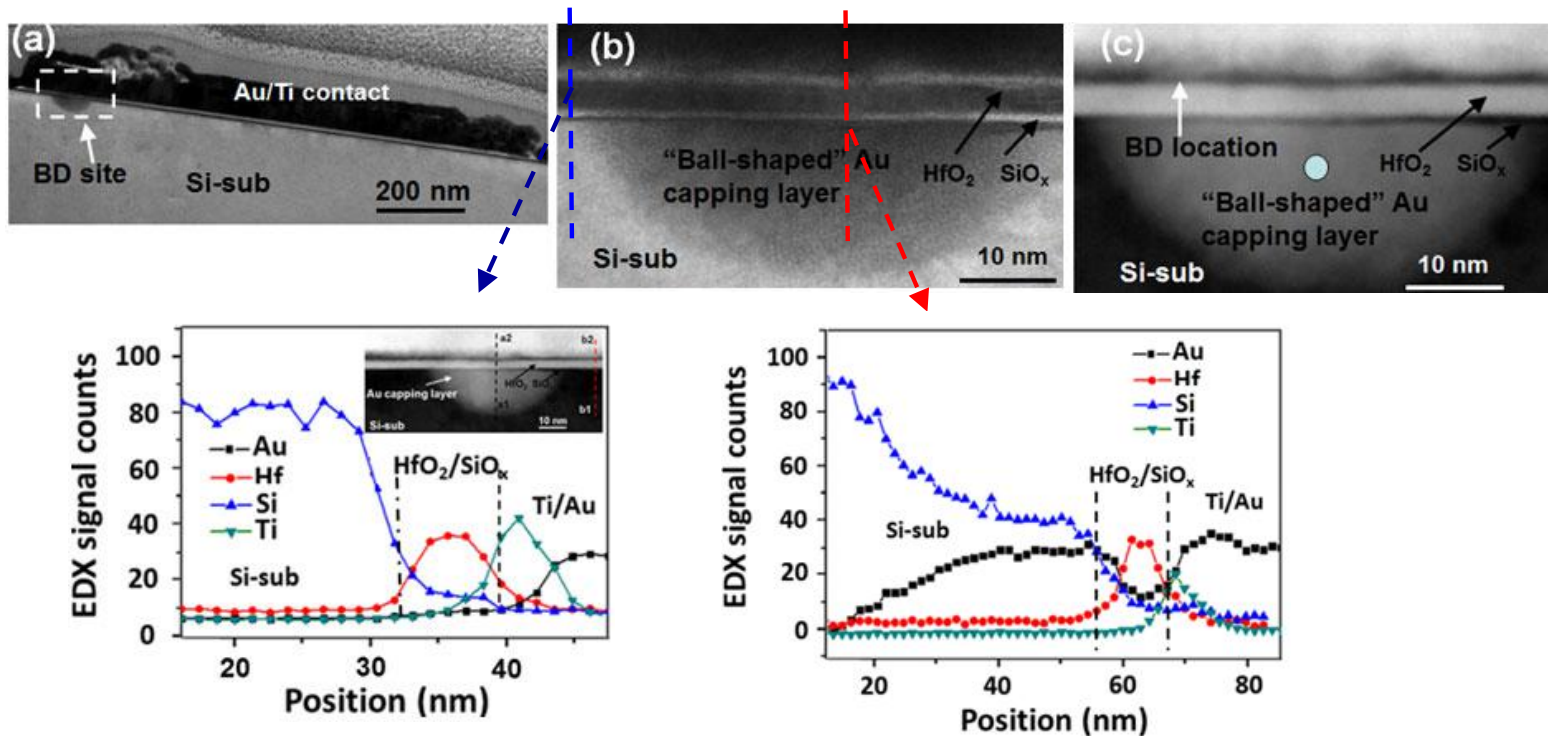
- ❖ No micro-structural changes of HK/IL visible
- ❖ Possible structural modification at the surface masked by the Pt protective layer.

Gate electrode region:

- ❖ Metal migration to Si-substrate through HK/IL stack

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Increased BD Hardness in HK/IL stacks



- Dielectric stack is still intact
- Migration of Au/Ti gate electrode material at the BD region-DBIM

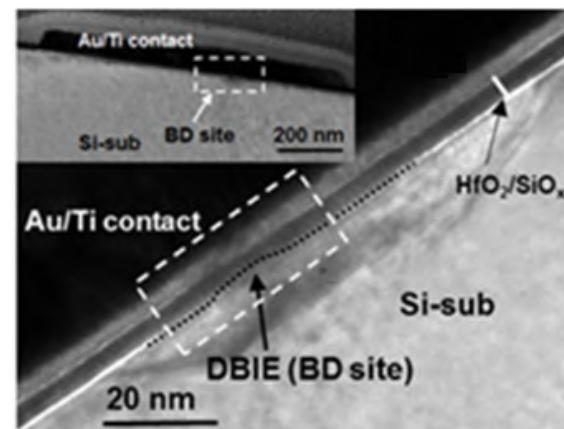
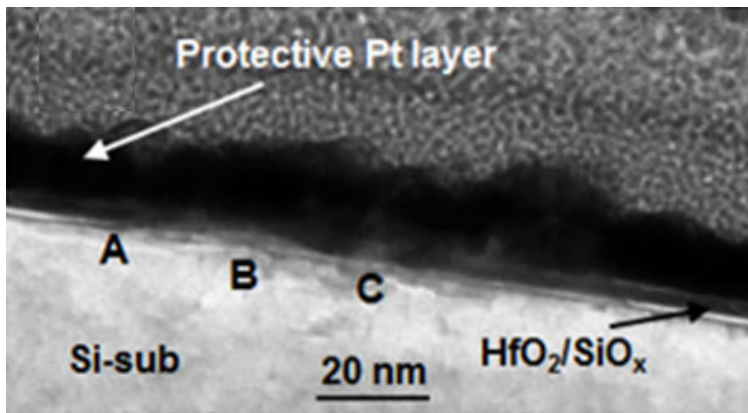
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Increased BD Hardness in HK/IL stacks (Substrate Injection)

Blanket region (B4)

Gate electrode region (G3)

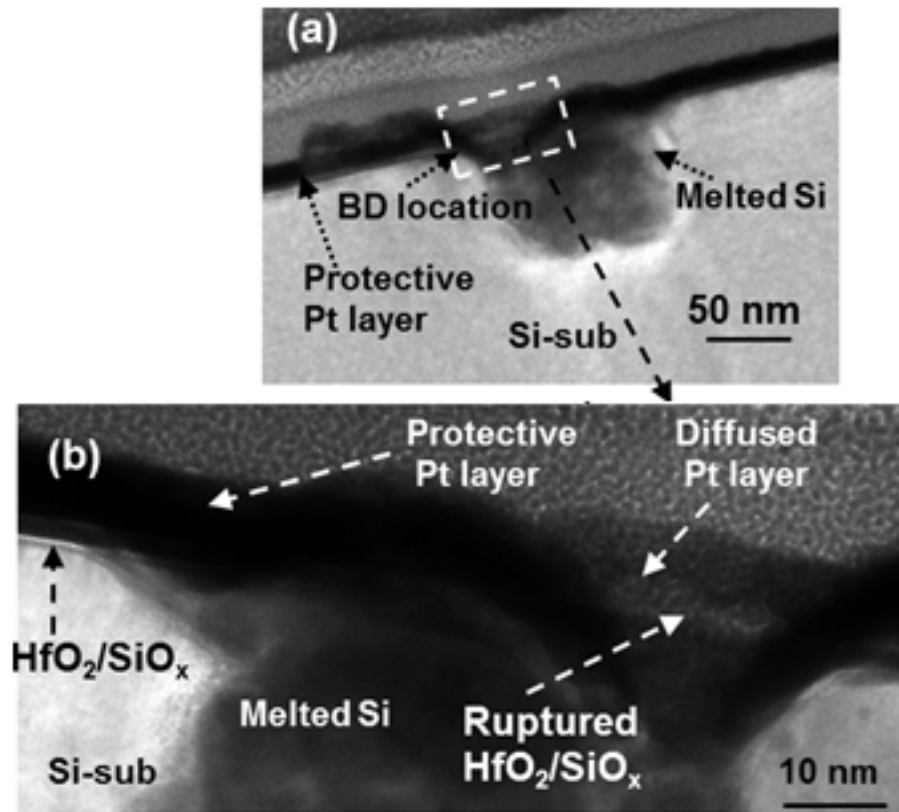


- Epitaxial Si hillocks extended from the Si substrate - DBIE at the BD spot- At both B4 and G3.
- SiO_x IL- ruptured
- Metal migration at the BD spot of Gate electrode region

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Catastrophic Failure (Substrate Injection)

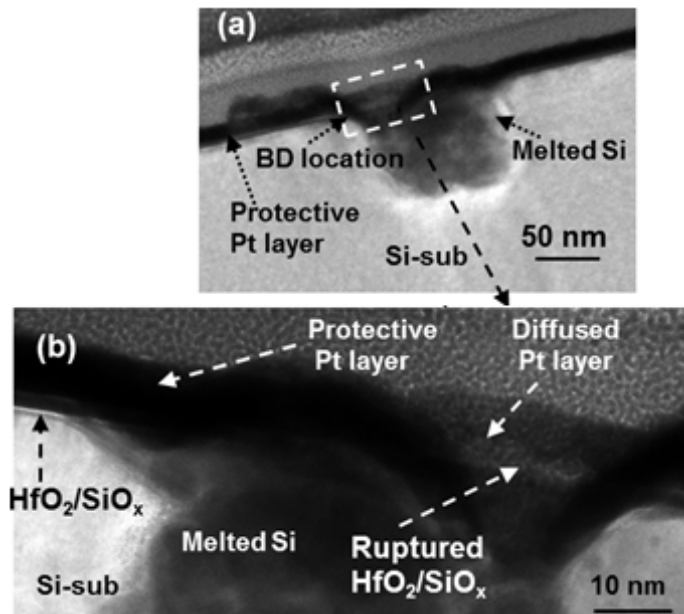
Blanket region



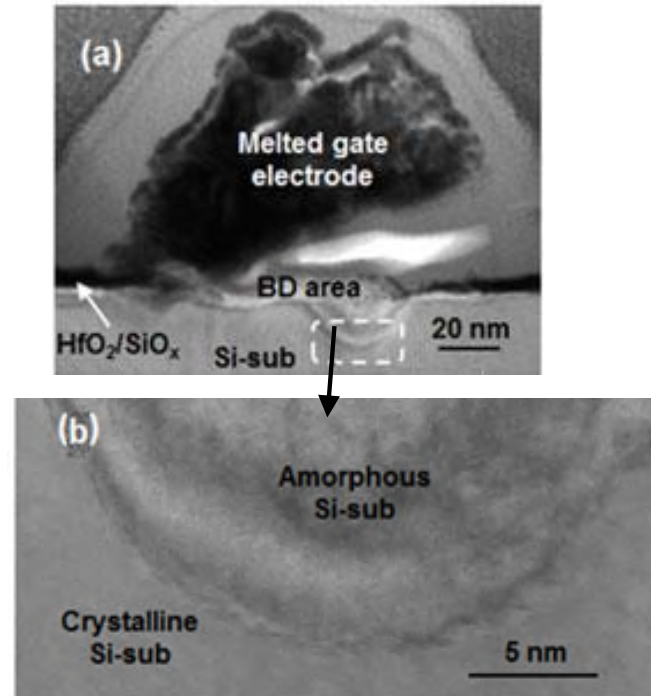
- Rupture of HfO₂/SiO_x stack
- Si-substrate melting
- Gate electrode melting and diffusion

Catastrophic Failure (Substrate Injection)

Blanket region



Gate electrode region



- Rupture of HfO₂/SiO_x stack
- Si-substrate melting
- Gate electrode melting and diffusion

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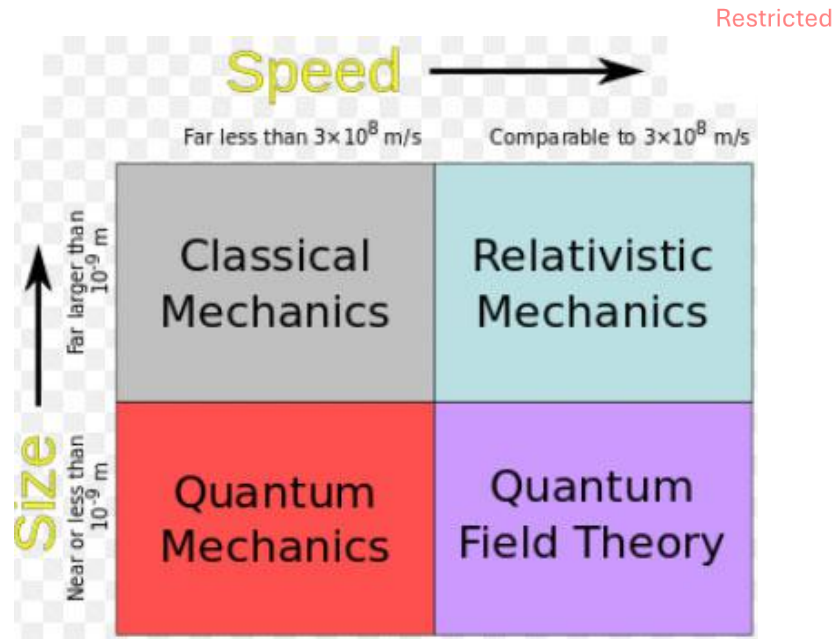
OPTIONAL
Only for Reading

Introduction to Quantum Mechanics and its relevance in Nano-MOSFETs

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Topics

- Relevance of quantum mechanics in Nanoscale devices and Nano-MOSFET
- Understanding of Electrostatics and transport in Nanoscale MOSFETS using quantum mechanics



Classical Mechanics deals with the motion of objects under forces or their own momentum. ... **Quantum Mechanics** deals with the behavior of objects at the microscopic level where matter and energy start to mathematically converge

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Principles of Quantum Mechanics

What is Quantum Mechanics?

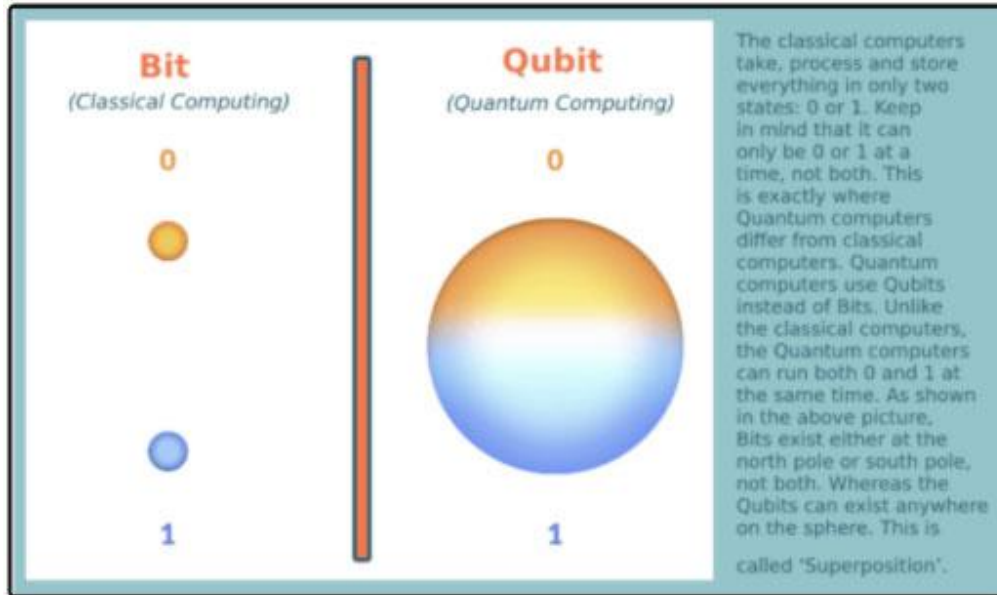
QM is the theory of the behavior of very small objects (e.g. molecules, atoms, nuclei, elementary particles, quantum fields, etc.)

Why *Quantum* Mechanics?

One of the essential differences between classical and quantum mechanics is that physical variables that can take on continuous values in classical mechanics (e.g. energy, angular momentum) can only take on discrete (or quantized) values in quantum mechanics (e.g. the energy levels of electrons in atoms, or the spins of elementary particles, etc).

Is quantum mechanics important/relevant?

- **Microelectronics** functions thanks to our understanding of basic quantum mechanics -> phones, computers ...
- Nuclear power is based on quantum mechanics ... and photovoltaics and thermoelectric too ... quantum mechanics can help solve **energy** problems
- New **materials** function following quantum mechanics: graphene, superconductors, semiconductors ... with vast current and potential applications
- More recently, people have been developing technologies based on quantum mechanics for **quantum telecommunication, quantum cryptography and quantum computing!**

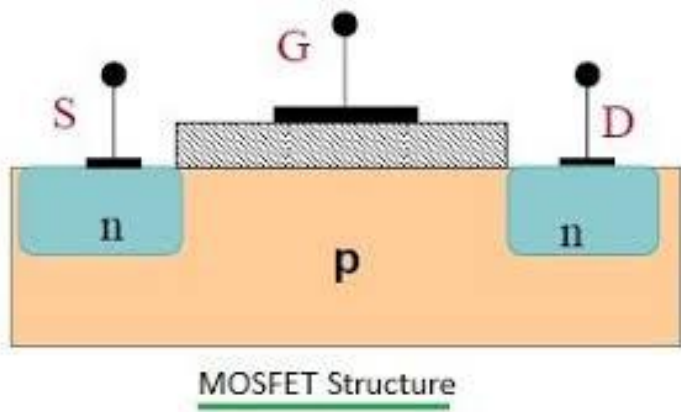


Nano-systems/Nanostructures

- Systems or structures/devices with dimension $<100\text{nm}$ at least in one dimension
- Low-dimensional systems
 - 2-Dimensional- Nanosheets, graphene, h-BN
 - 1-Dimensional- Nanotube, Nanowire
 - 0-Dimensional- Quantum dots

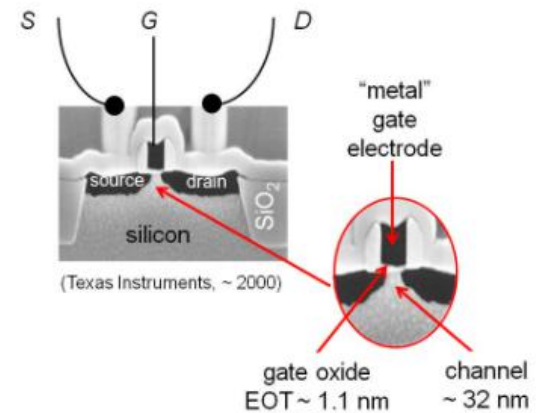
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Scaling of MOSFETS and its impact on working principle



Scaling

Classical physics

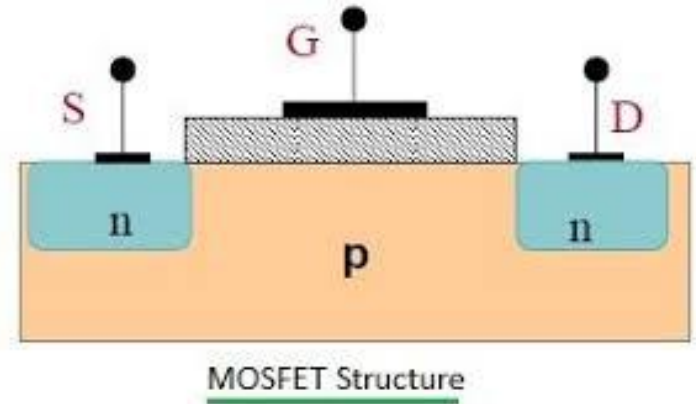


Quantum mechanical effects?

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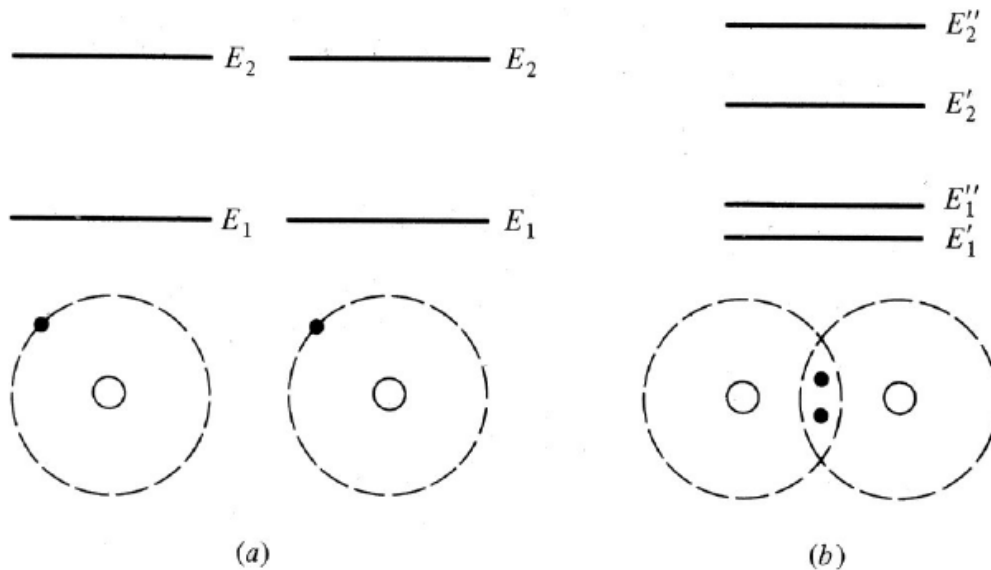
Nanoscale MOSFETs:

- Impact of quantum mechanical effect
 - Electrostatics
 - Carrier transport
- Density of states
 - Discrete energy band, confinement (1-D, 2-D, 3-D), Quantum well
- Quantum tunneling
- Scattering and Ballistic transport



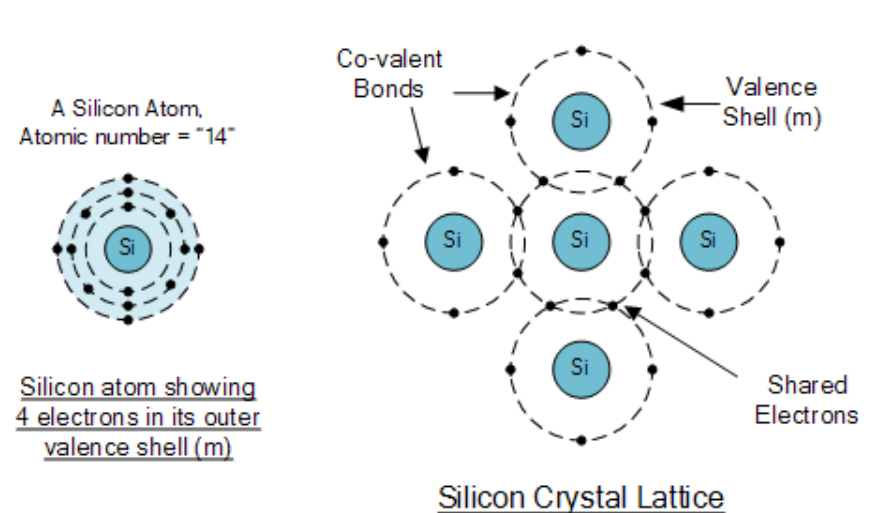
Discrete Energy band at Nanoscale

From quantum physics that electrons in an atom can occupy only discrete energy levels.



Two hydrogen atoms: (a) noninteracting and (b) interacting.

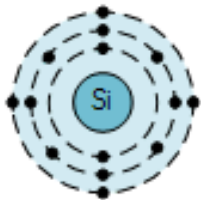
Splitting of energy levels is illustrated for (b).



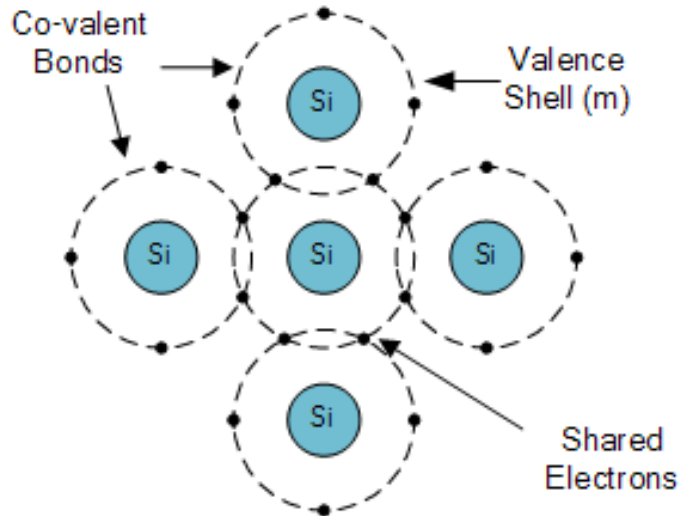
Discrete Energy band at Nanoscale

From quantum physics that electrons in an atom can occupy only discrete energy levels.

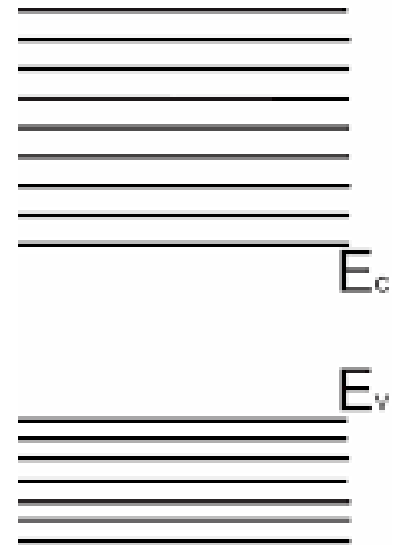
A Silicon Atom,
Atomic number = "14"



Silicon atom showing
4 electrons in its outer
valence shell (m)

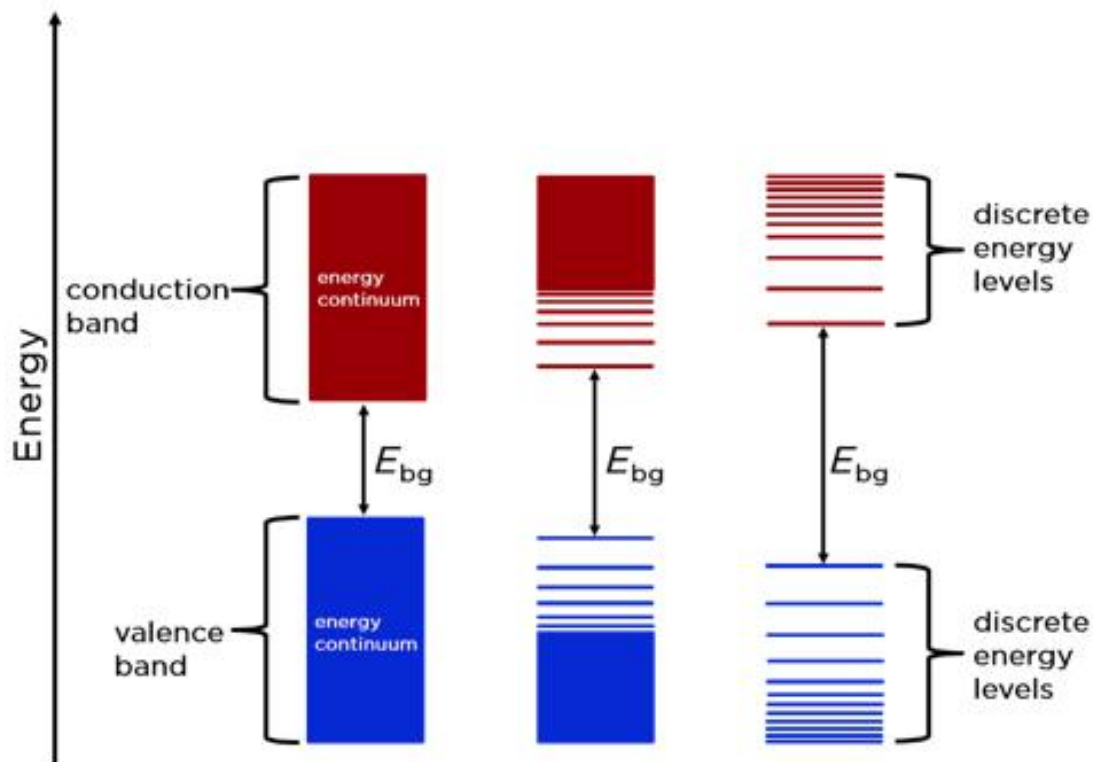


Silicon Crystal Lattice



Discrete Energy band at Nanoscale

From quantum physics that electrons in an atom can occupy only discrete energy levels.



- Leads increase in the energy bandgap
- Results in confinement of the charges

Quantum confinement

- Trap particles and restrict their motion
- Quantum confinement produces new material behavior/phenomena
- “Engineer confinement”- control for specific applications
- Structures
 - Quantum dots (0-D) only confined states, and no freely moving ones
 - Nanowires (1-D) particles travel only along the wire
 - Quantum wells (2-D) confines particles within a thin layer

<http://www.me.berkeley.edu/nti/englander1.ppt>
<http://phys.educ.ksu.edu/vqm/index.html>

(Scientific American)

Quantum confinement

Restricted

no confinement

bulk
(3D materials)

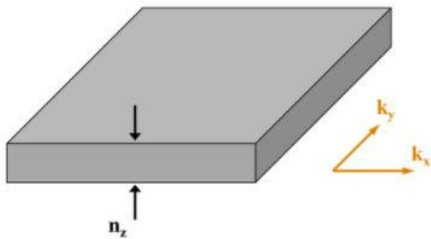
Quantum Wells = “2 dimensional” structures

Quantum Wires = “1 dimensional” structures

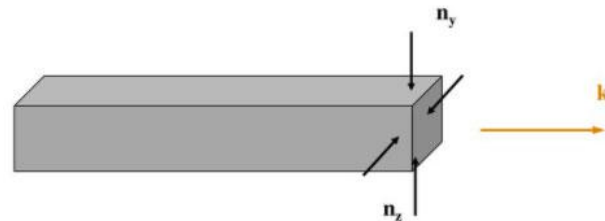
Quantum Dots = “0 dimensional” structures!!



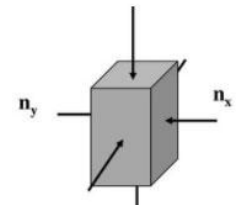
There is 1 Dimensional Quantization



There is 2 Dimensional Quantization



There is 3 Dimensional
Quantization

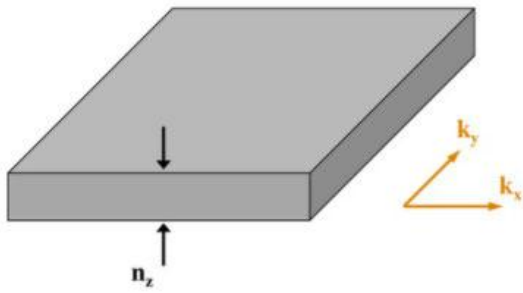


Restricted

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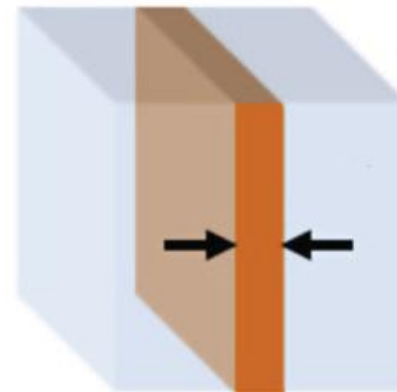
Quantum confinement

There is 1 Dimensional Quantization



- 2-D materials
- Graphene, h-BN, ..

1D confinement
Q-well/ultrathin film
(2D materials)

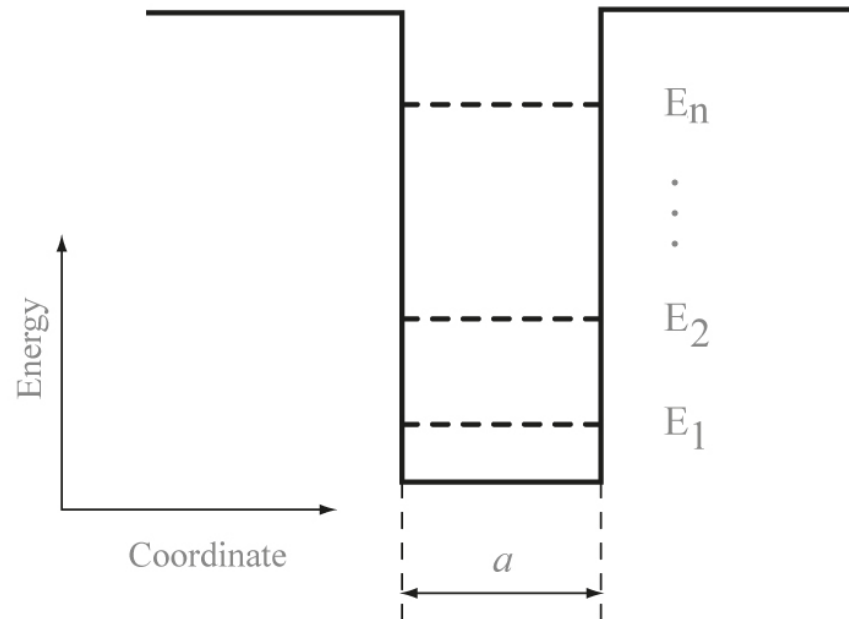
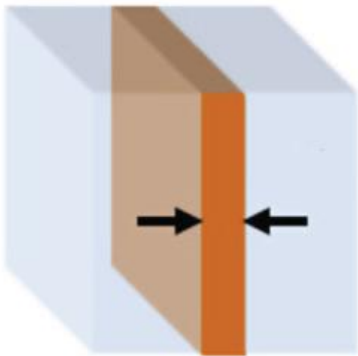


Restricted

Restricted

Quantum well (Rectangular)

1D confinement
Q-well/ultrathin film
(2D materials)



<https://www.youtube.com/watch?v=IkZpUAeVkik&t=186s>

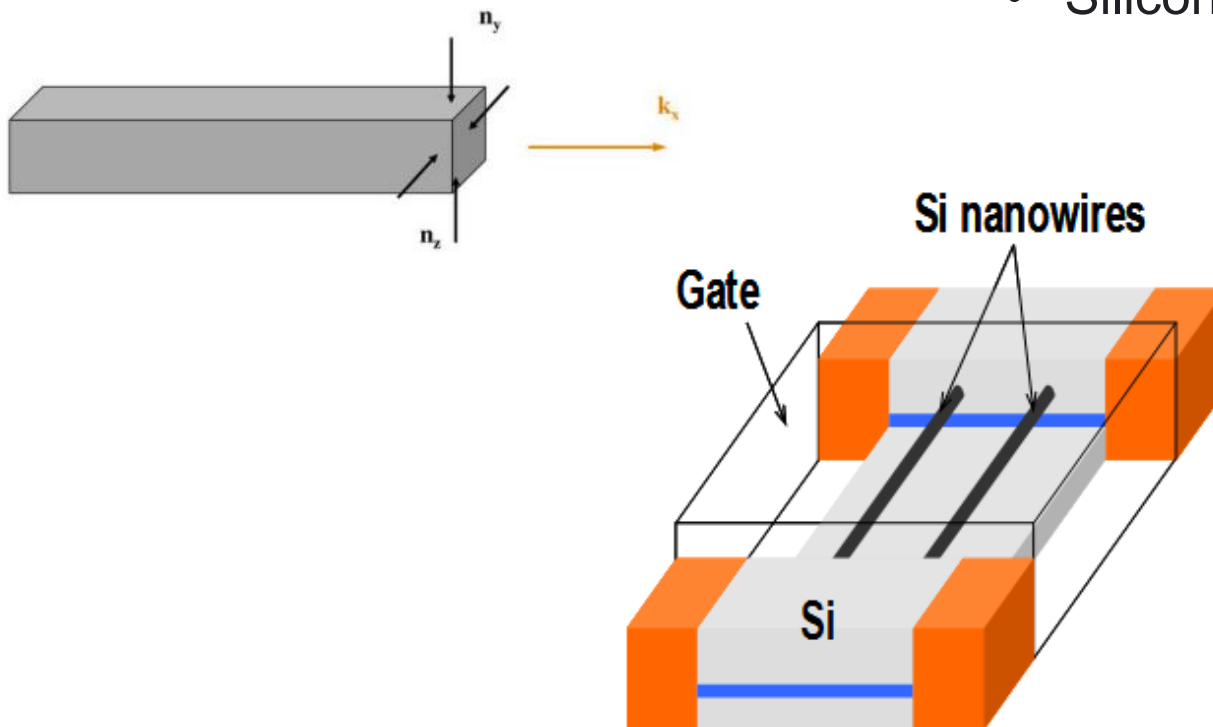
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Quantum confinement(2-D)

There is 2 Dimensional Quantization

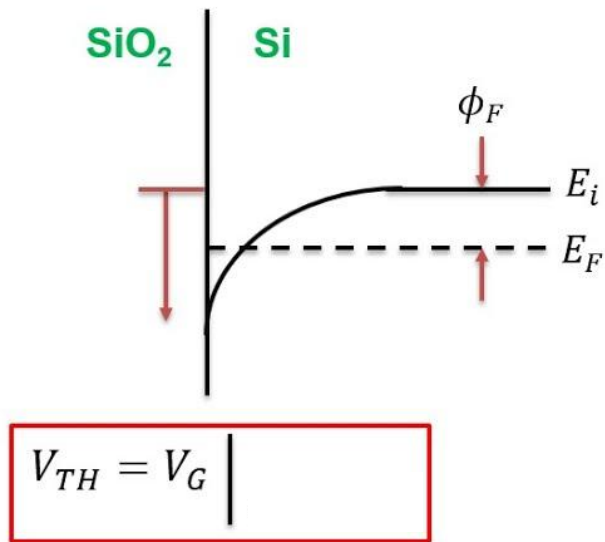
- Silicon nanowire MOSFET



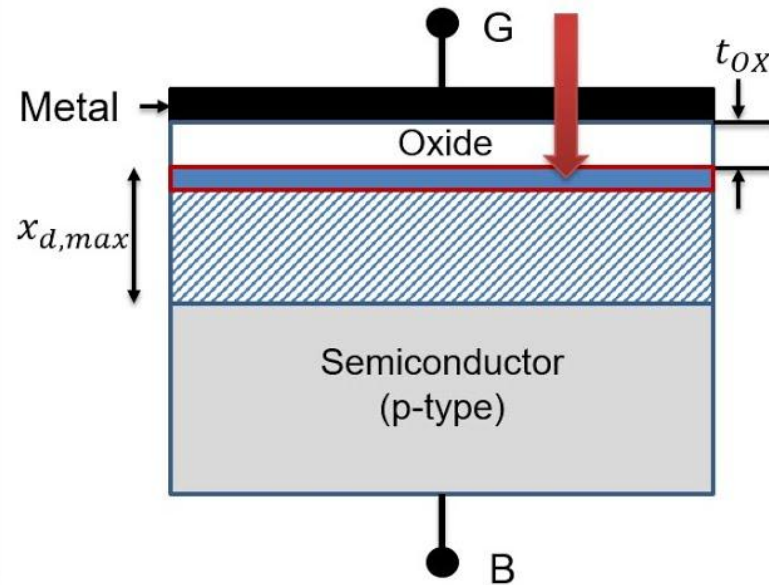
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Triangular potential barrier (Inversion charge density)

Restricted



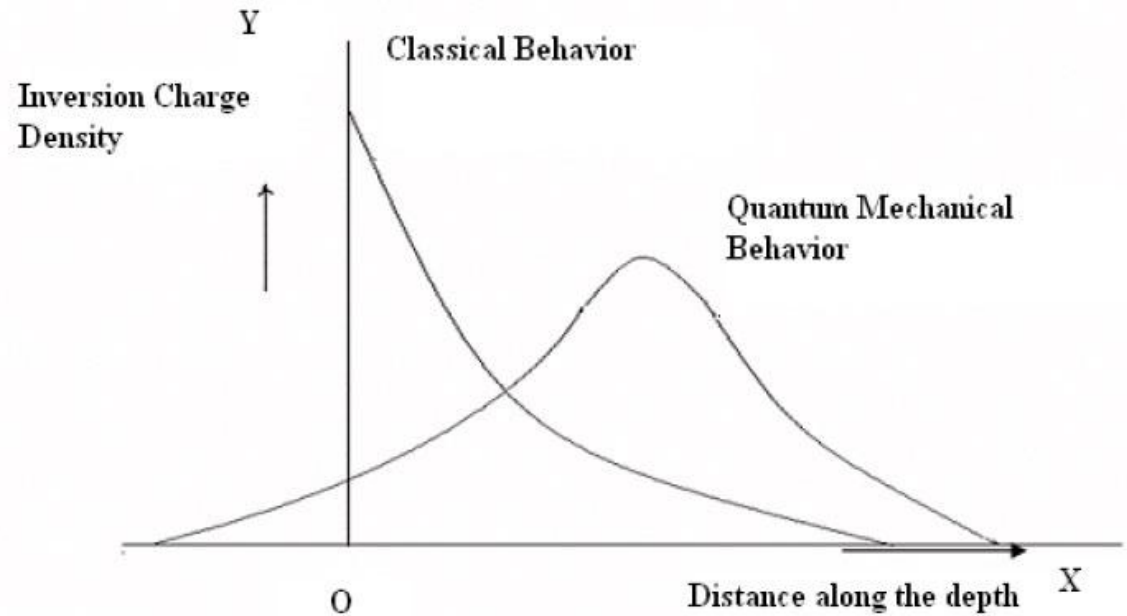
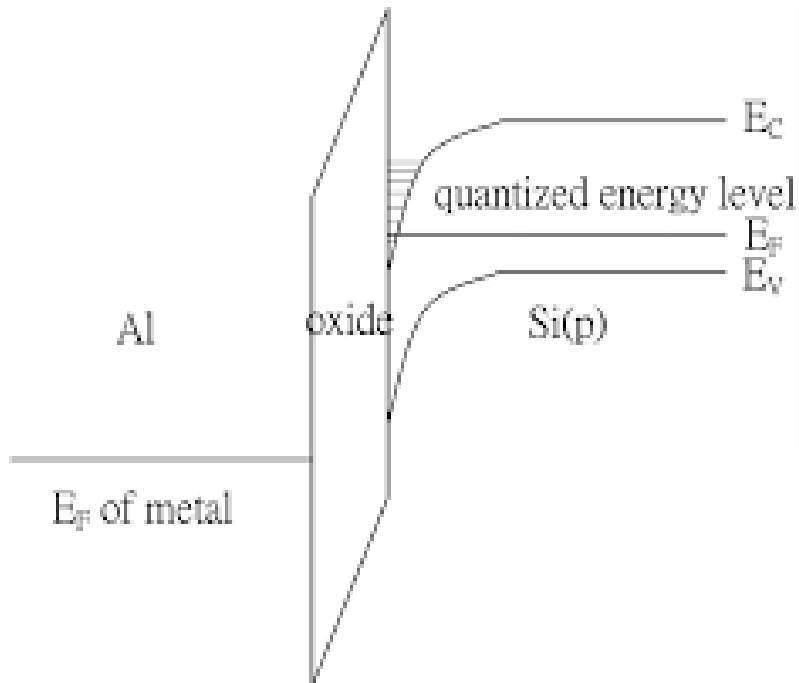
Inversion charge



Restricted

Triangular potential barrier (Inversion charge density)

Restricted



Electron Concentration Distribution in the Silicon Substrate in Classical and Quantum Mechanical Cases.

Restricted

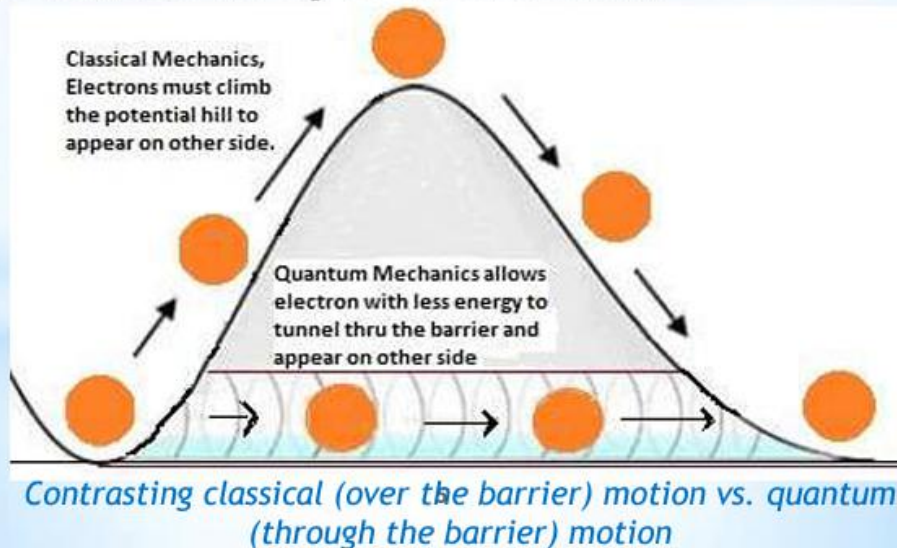
Quantum Mechanical Tunneling

Quantum tunnelling is not predicted by the laws of classical mechanics where surmounting a potential barrier requires enough potential Energy.

It has important applications in the tunnel diode, quantum computing, and in the scanning tunnelling microscope. The effect was predicted in the early 20th century, and its acceptance as a general physical phenomenon came mid-century.

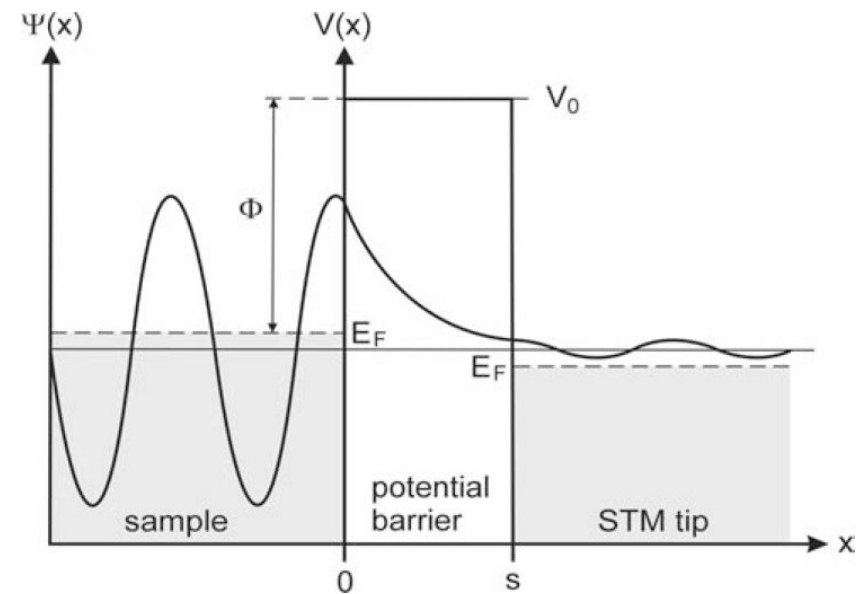
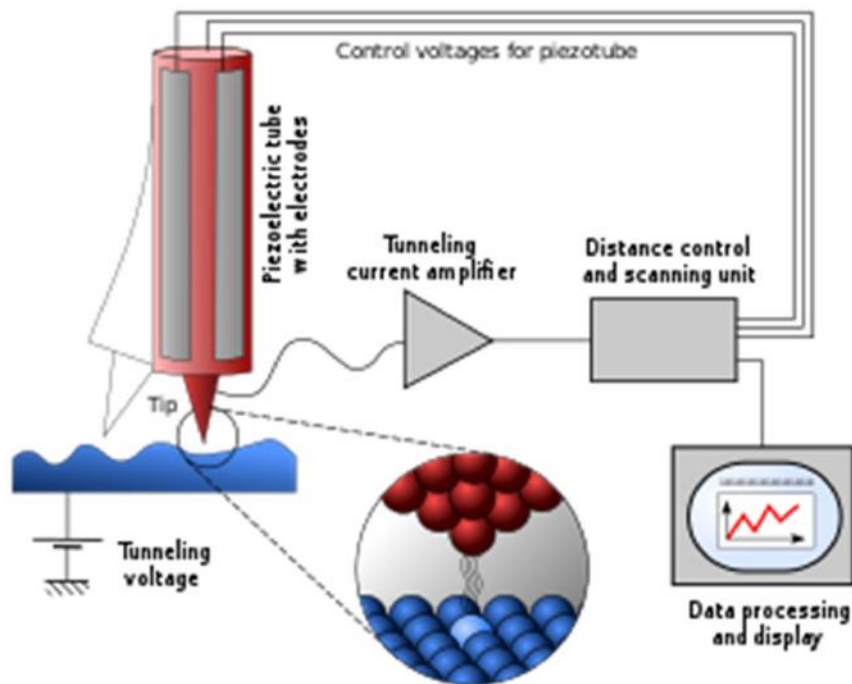
Quantum Mechanical Tunneling

Violating Classical Mechanics (Cont.): However, If the ball does not have enough kinetic energy ($E < V$), to overcome the barrier it will never roll from one valley to the other. In contrast, when quantum effects are taken into effect, the ball can "tunnel" through the barrier to the other valley, even if its kinetic energy is less than the potential energy of the barrier to the top of one of the hills.



Quantum Mechanical Tunneling

Scanning tunneling microscope (1981)



Introduction to quantum mechanics

Plenty of Room at the Bottom

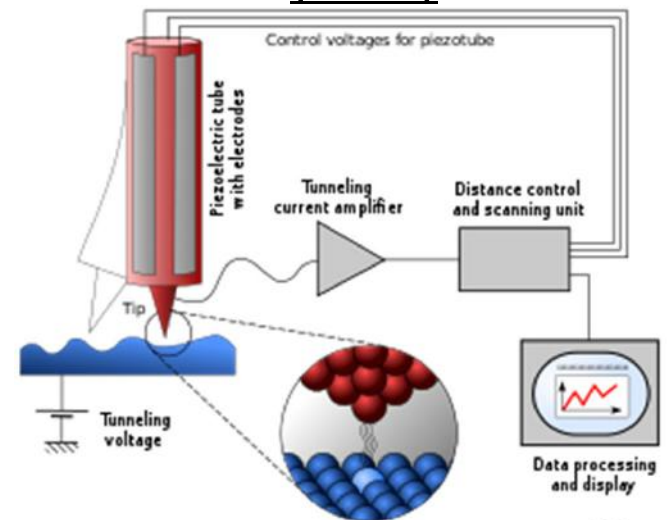
Richard P. Feynman
(Dated: Dec. 1959)

This is the transcript of a talk presented by Richard P. Feynman to the American Physical Society in Pasadena on December 1959, which explores the immense possibilities afforded by miniaturization.

I imagine experimental physicists must often look with envy at men like Kamerlingh Onnes, who discovered a field like low temperature, which seems to be bottomless and in which one can go down and down. Such a man is then a leader and has some temporary monopoly in a scientific adventure. Percy Bridgman, in designing a way to obtain higher pressures, opened up another new field and was able to move into it and to lead us all along. The development of ever higher vacuum was a continuing development of the same kind.

Furthermore, it can be read if it is so written. Let's imagine that it is written in raised letters of metal; that is, where the black is in the Encyclopedia, we have raised letters of metal that are actually $1/25,000$ of their ordinary size. How would we read it?

Scanning tunneling microscope (1981)



Quantum well-LASER

- A laser is created when electrons in the atoms in optical materials like glass, crystal, or gas absorb the energy from an electrical current or a light. That extra energy “excites” the electrons enough to move from a lower-energy orbit to a higher-energy orbit around the atom's nucleus.

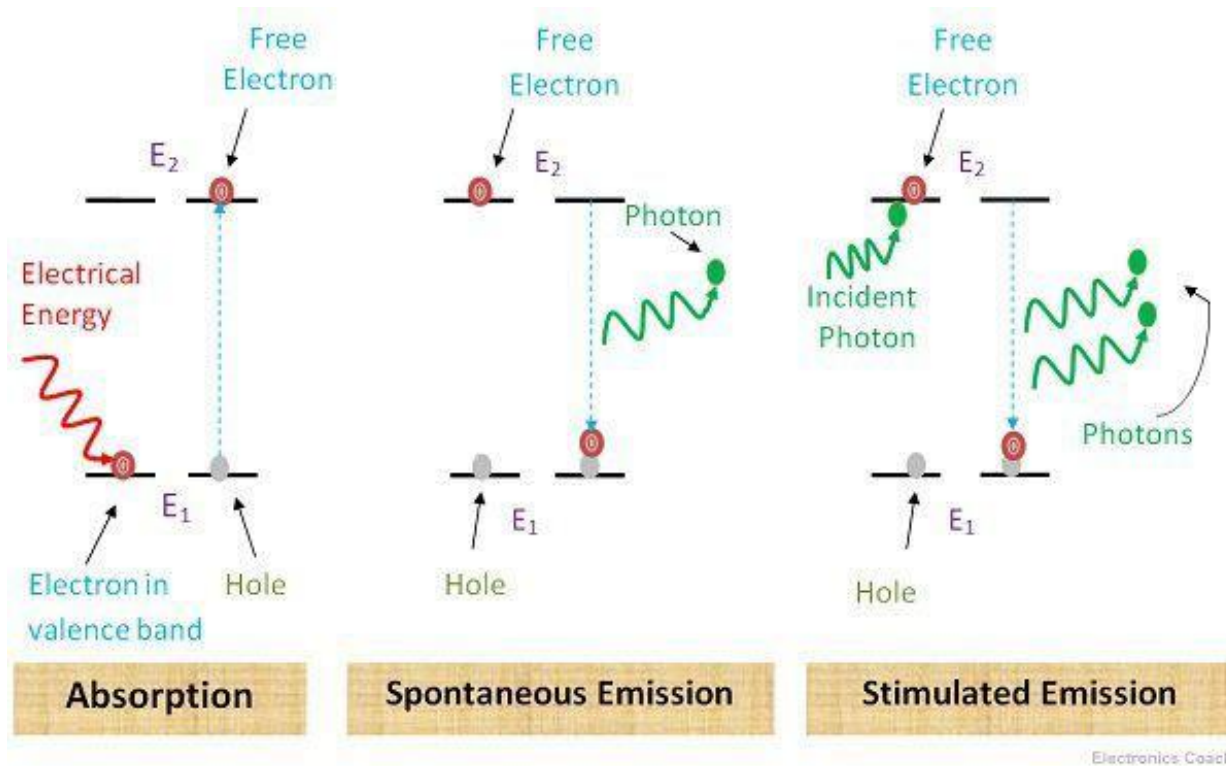
THE LASER

All the animations and explanations on
www.toutestquantique.fr

https://www.youtube.com/watch?v=R_QOWbkc7UI&t=50s

Quantum well-LASER

- A laser is created when electrons in the atoms in optical materials like glass, crystal, or gas absorb the energy from an electrical current or a light. That extra energy “excites” the electrons enough to move from a lower-energy orbit to a higher-energy orbit around the atom's nucleus.



Quantum well-LASER

- The much shorter wavelengths can be obtained from quantum well lasers than from conventional laser diodes using a particular semiconductor material. The efficiency of a quantum well laser is also greater than a conventional laser diode due to the stepwise form of its density of states function.



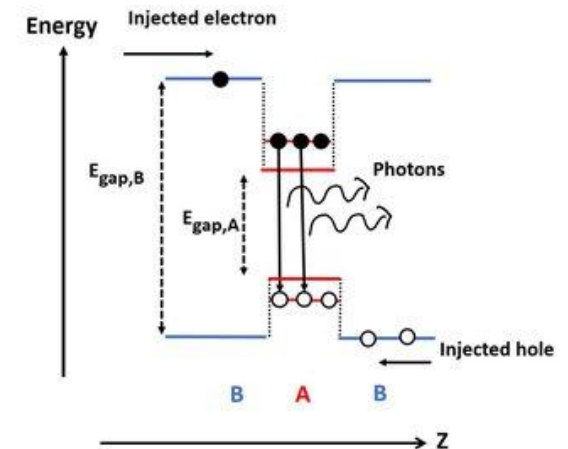
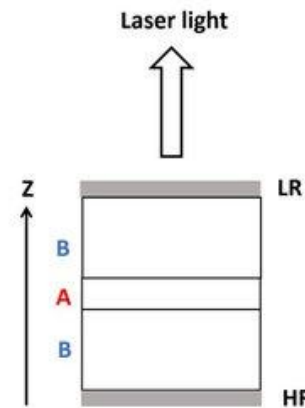
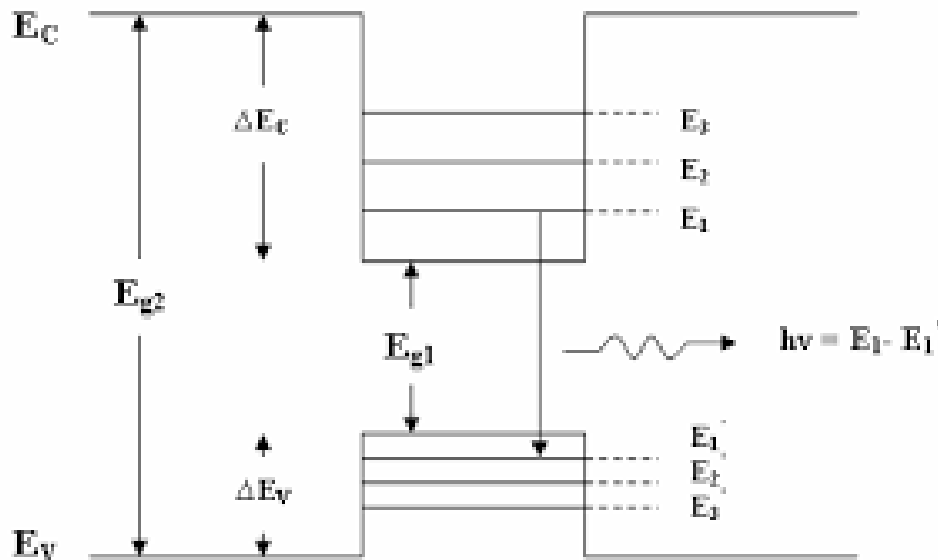
$$\lambda = \frac{hc}{\Delta E}$$

https://www.youtube.com/watch?v=RQUFOA_8LZg&t=29s

Restricted

Quantum well-LASER

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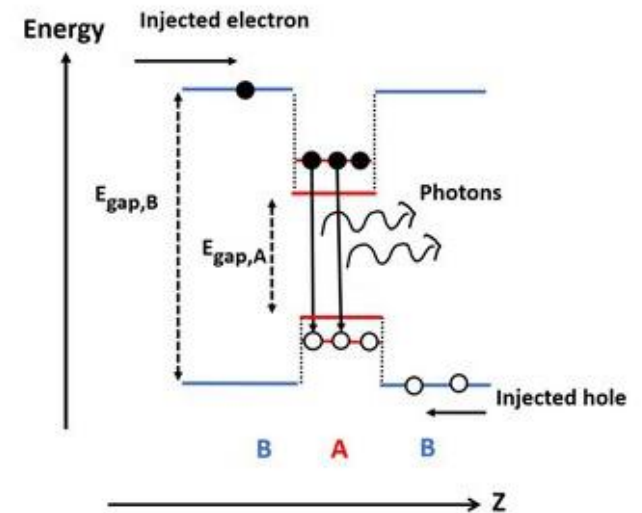
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Quantum well-LASER

- Wavelength relation to transition energy

$$\lambda = \frac{hc}{\Delta E}$$

- ΔE is the transition of energy levels



Quantum well lasers have attracted a great deal of attention by their many advantages such as

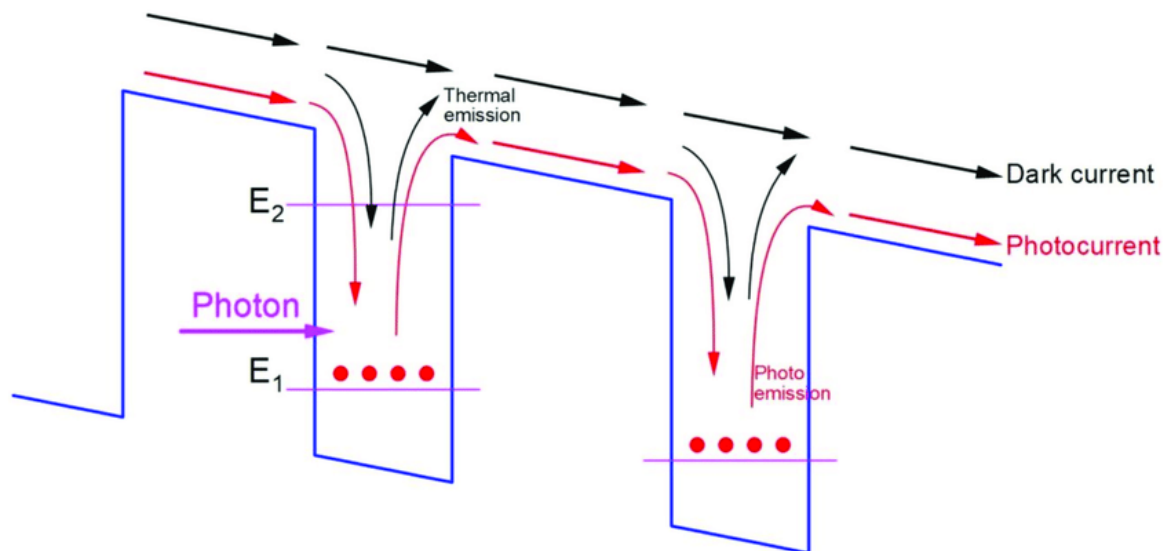
- Low threshold current density,**
- Excellent temperature feature, high modulation rate and**
- Wavelength adjustability**

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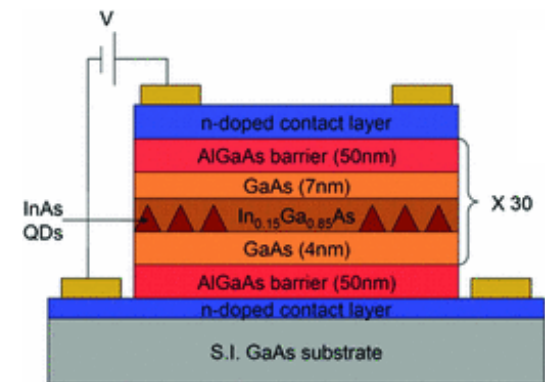
Quantum well-Photodetector

Quantum wells can also be explored to work as a photodetector

The quantum well photodetector is a semiconductor photon detector relying on inter-subband absorption within either the conduction band (n-type) or the valence band (p-type).



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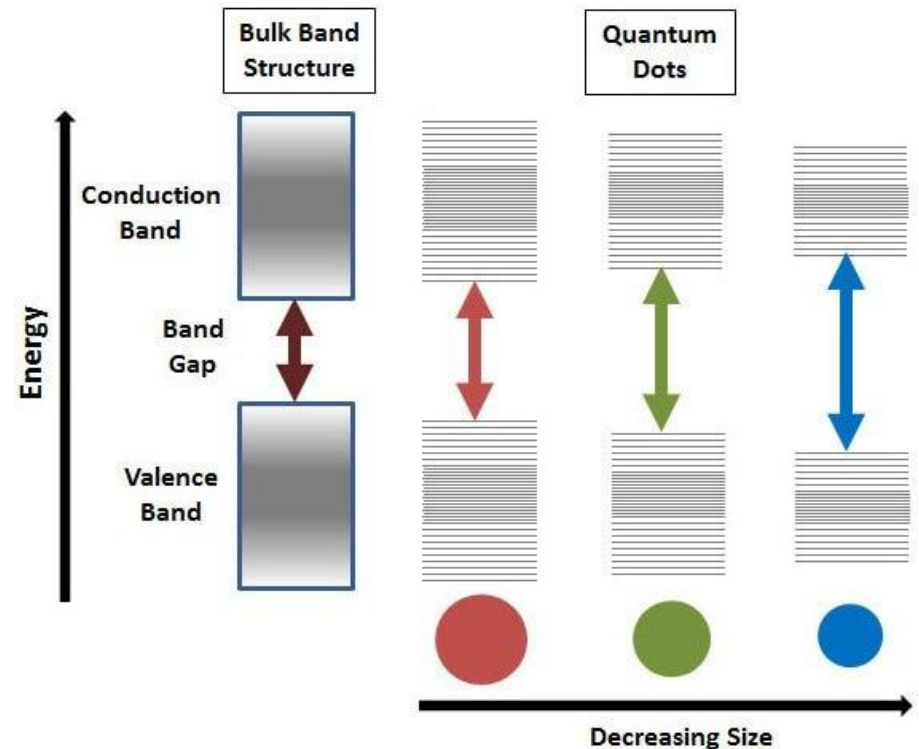


Quantum Dots- Display

Quantum dots are nanocrystals made of semiconductor materials, typically nanometers in size.

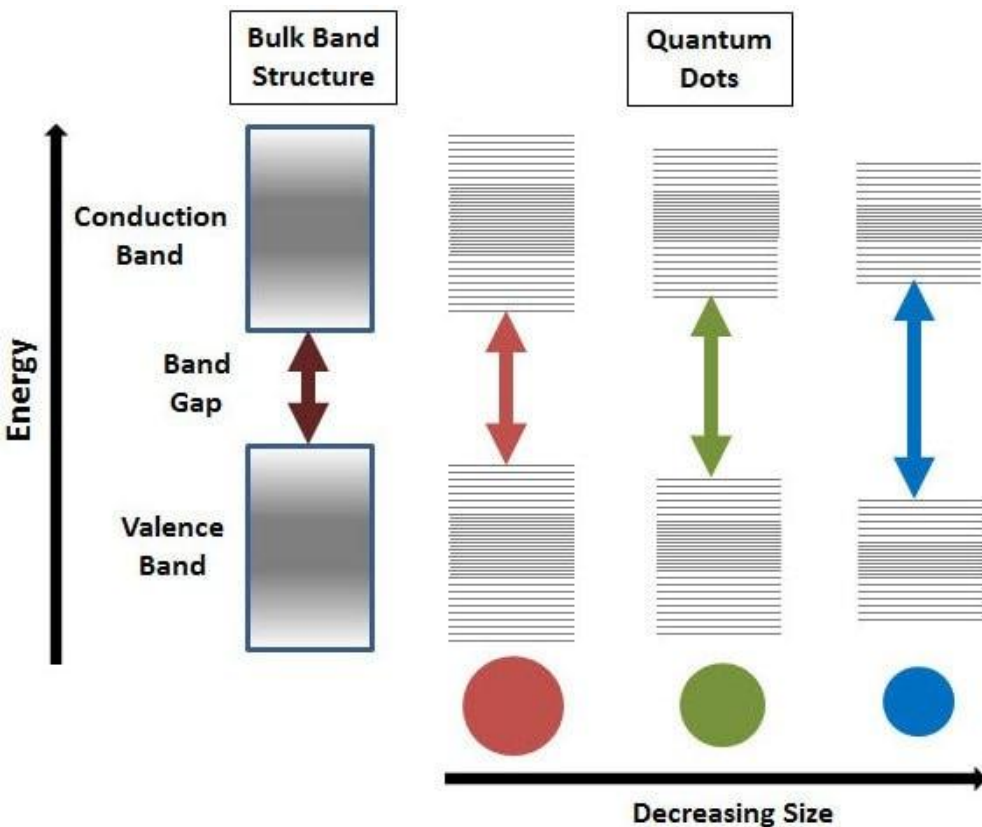
- Their unique property is that they emit light of specific wavelengths (colors) when exposed to light or electricity, and the wavelength depends on their size:

- Smaller dots emit blue light.
- Larger dots emit red light.



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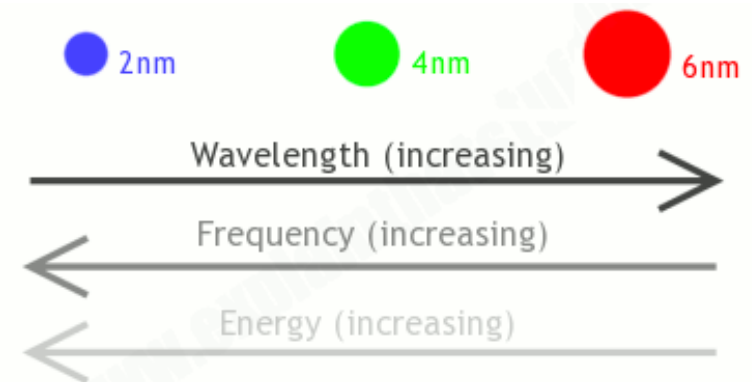
Quantum Dots- Display



- Wavelength relation to transition energy

$$\lambda = \frac{hc}{\Delta E}$$

- ΔE is the transition of energy levels



www.explainthatstuff.com

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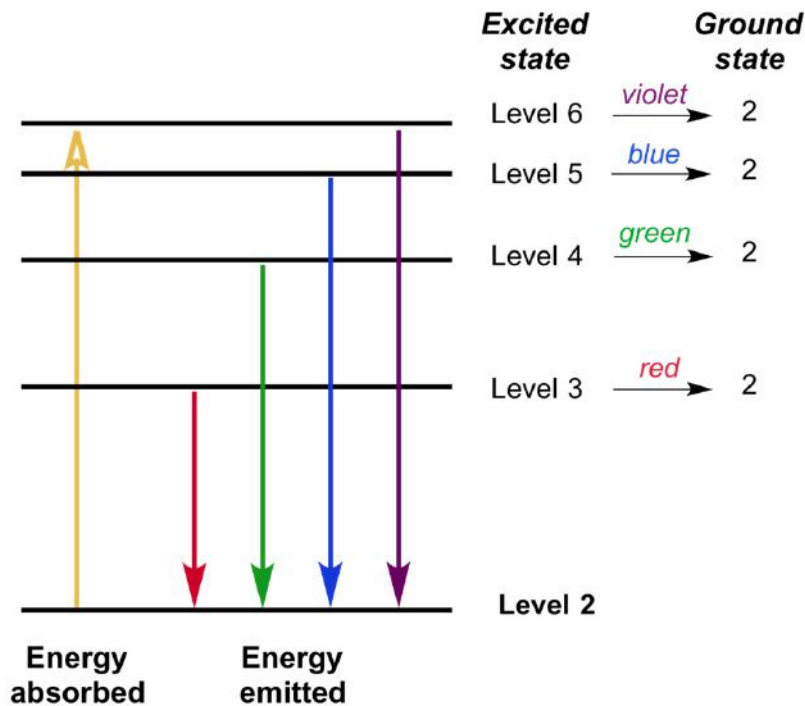
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Quantum Dots-Display technology

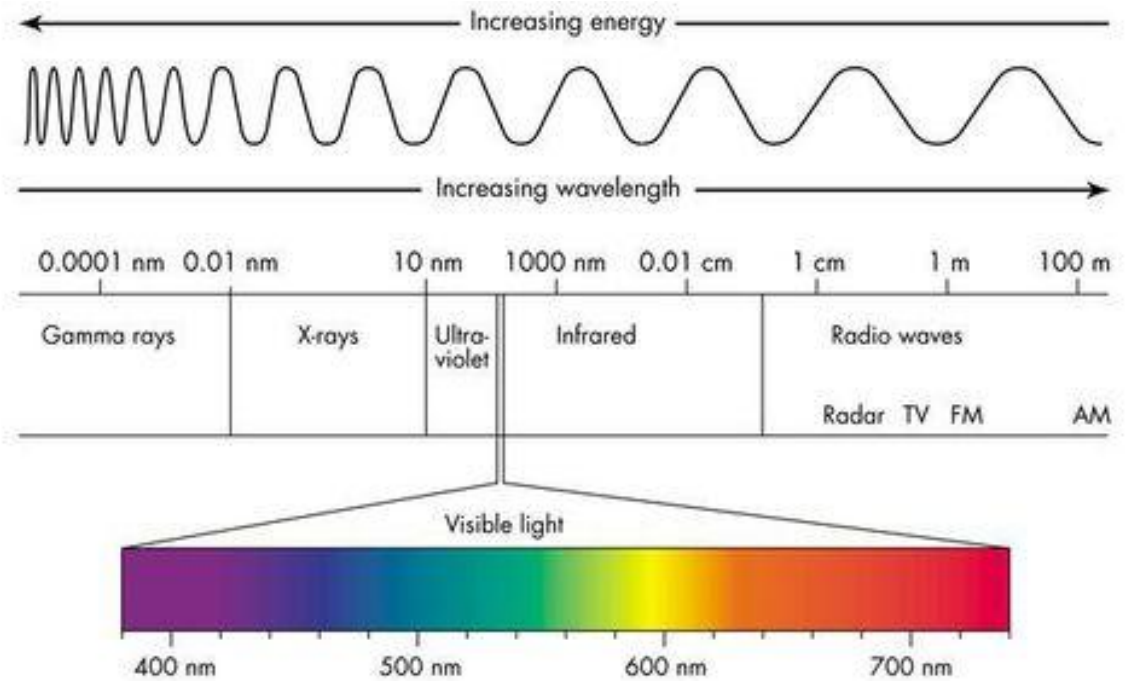
- Wavelength relation to transition energy

$$\lambda = \frac{hc}{\Delta E}$$

- ΔE is the transition of energy levels



$$\Delta E = \frac{hc}{\lambda}$$



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Quantum Dots-Display technology

Working of Quantum Dot Displays:

1. Backlighting:

- In QD-enhanced displays (like QLED), a blue LED backlight is commonly used.

2. Quantum Dot Layer:

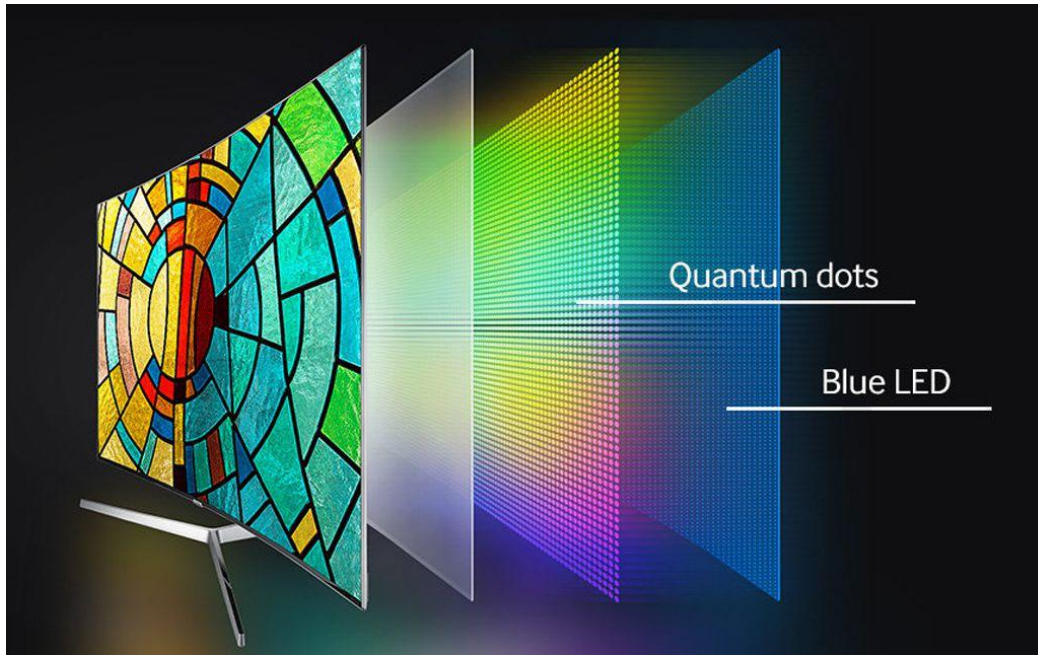
- A layer of quantum dots is placed between the backlight and the display panel. When the blue light hits the quantum dots:
 - Some quantum dots convert the blue light into highly pure red and green light.
 - The rest of the blue light passes through directly.

3. Color Filters:

- The emitted red, green, and blue light passes through a color filter to create the full spectrum of colors displayed on the screen.

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Quantum Dots-Display technology



<https://growthmarketreports.com/blog/what-is-the-quantum-dot-displays>

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Quantum Dots-Display technology



<https://www.youtube.com/watch?v=saoos9dzPUg>